DATA COMMUNICATION, COMPUTER NETWORK AND INTERNET BASICS

UNIT-09

What is Data Communications?

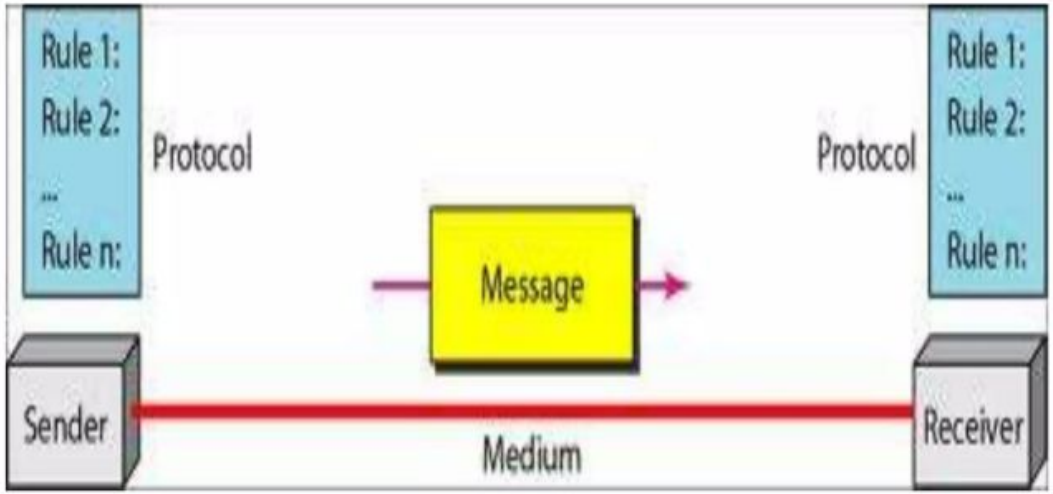
Exchange of data between two devices via some forms of transmission medium(such as wire cable) is Data Communications.

For data communications to occur, the communicating devices must be part of a communication system made of a combination of hardware and software.

The effectiveness of a data communication system depends on four fundamental characteristics:- delivery, accuracy, timeliness and jitter.

Components of Data Communication:

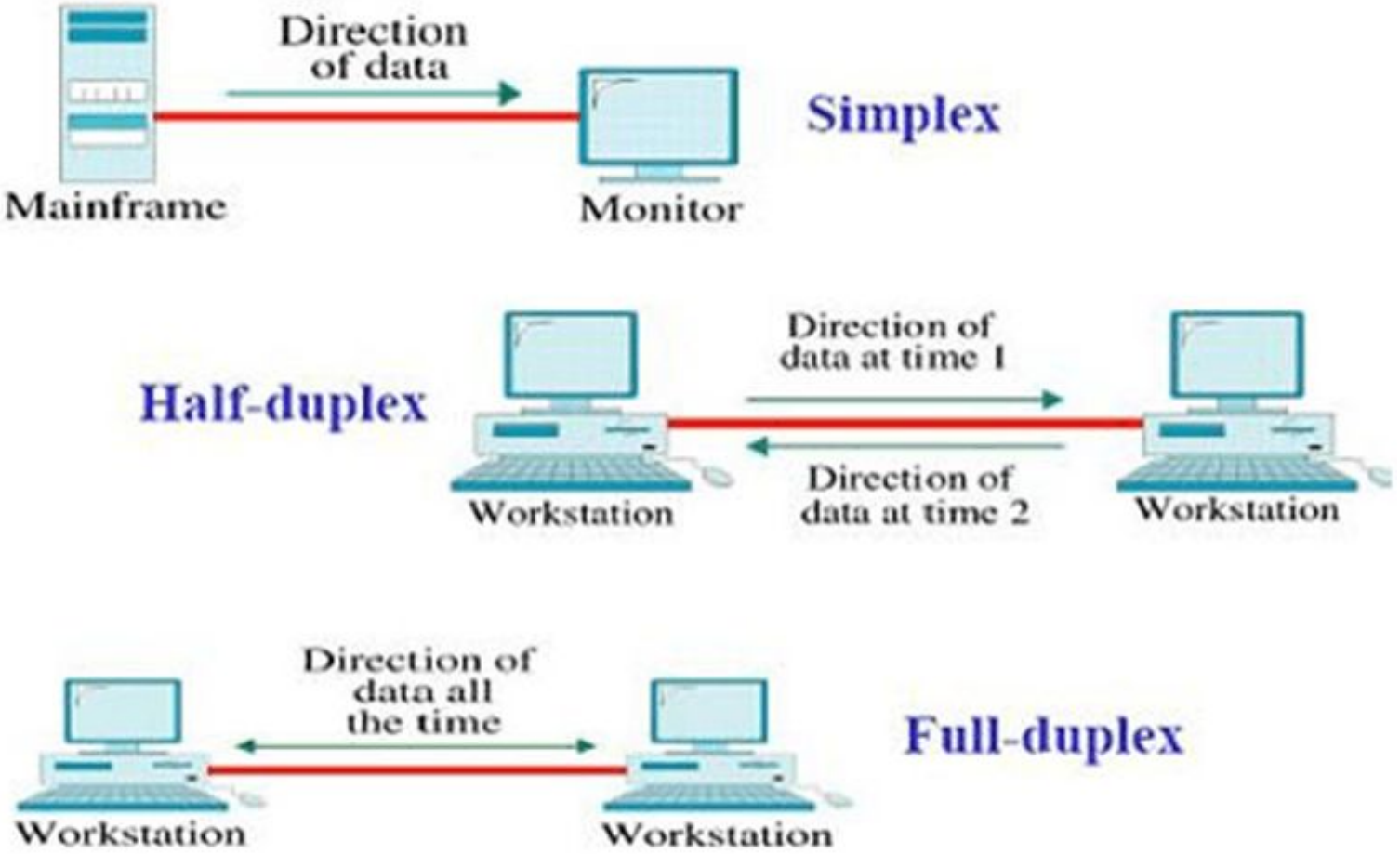
1.Sender 2.Receiver 3.Message 4.Tramsmission Medium
5. Protocol



The five components of data communication are:

1. **Message** - It is the information to be communicated. Popular forms of information include text, pictures, audio, video etc.
2. **Sender** - It is the device which sends the data messages. It can be a computer, workstation, telephone handset etc.
3. **Receiver** - It is the device which receives the data messages. It can be a computer, workstation, telephone handset etc.
4. **Transmission Medium** - It is the physical path by which a message travels from sender to receiver. Some examples include twisted-pair wire, coaxial cable, radio waves etc.
5. **Protocol** - It is a set of rules that governs the data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating.

Transmission Mode



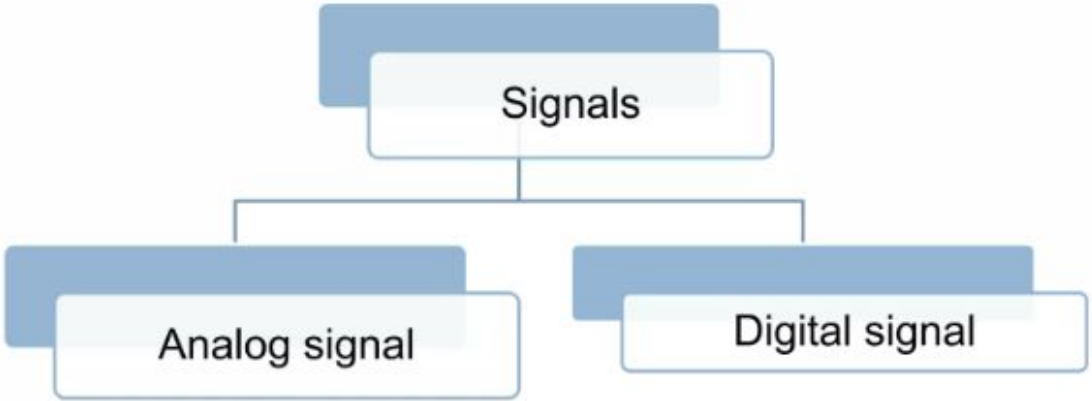
Data flow can occur in three ways:

1. **Simplex:** In simplex mode,the communication is unidirectional.only one of the devices on a link can transmit, the other can only receive. e.g. keyboards,monitors,etc.
2. **Half-duplex:** In this mode,each station can both transmit and receive,but not at the same time.When one device is sending,the other can only receive,and vice-versa. e.g. walkie-talkies,CB(citizens band) etc.

3. **Full Duplex :** In full duplex mode, both stations can transmit and receive simultaneously. One common example of full duplex is the Telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time. The full-duplex mode is used when communication in both directions is required all the time.

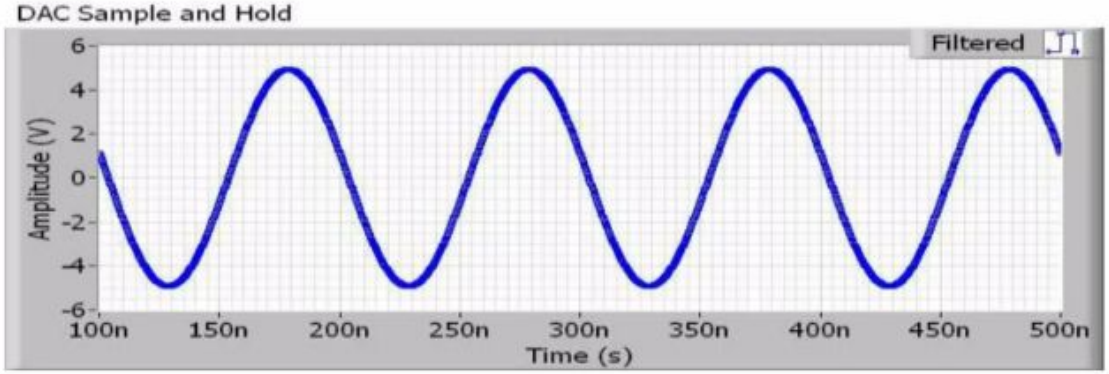
Signals

- There are two types of signals to transfer data.



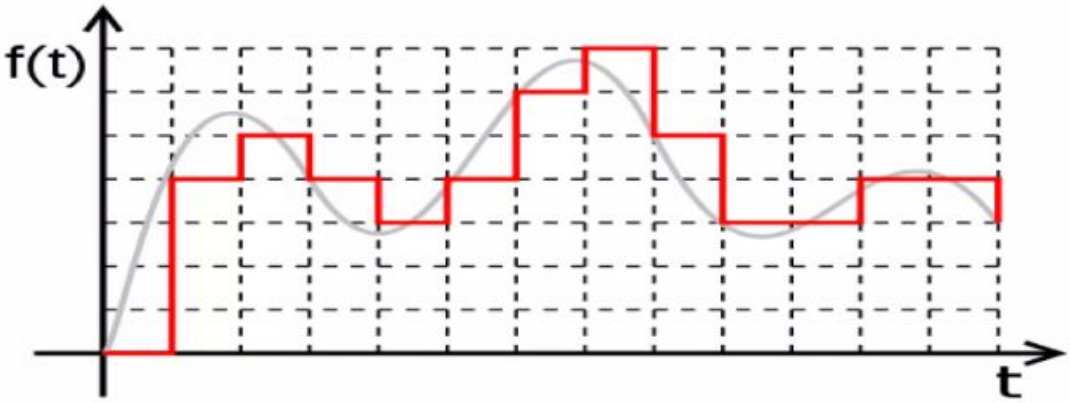
Analog Signals

An analog signal are continuous and passes through or includes an infinite number of continuous values along its path. The curve representing the analog signal passes through an infinite number of points.



Digital Signals:

A digital signal can have only a limited number of defined values. Although each value can be any number,it is as simple as 1 and 0.



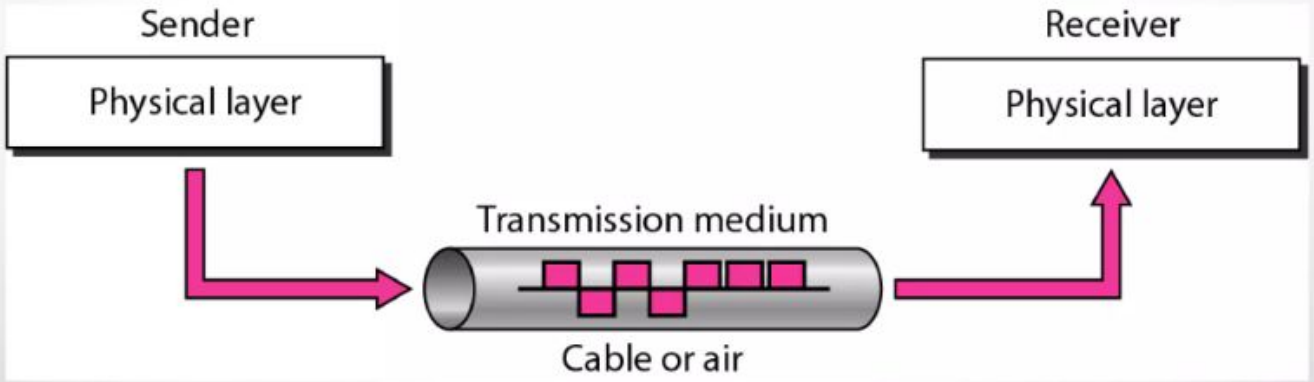
What is Tranmission Media ?

In data communication,

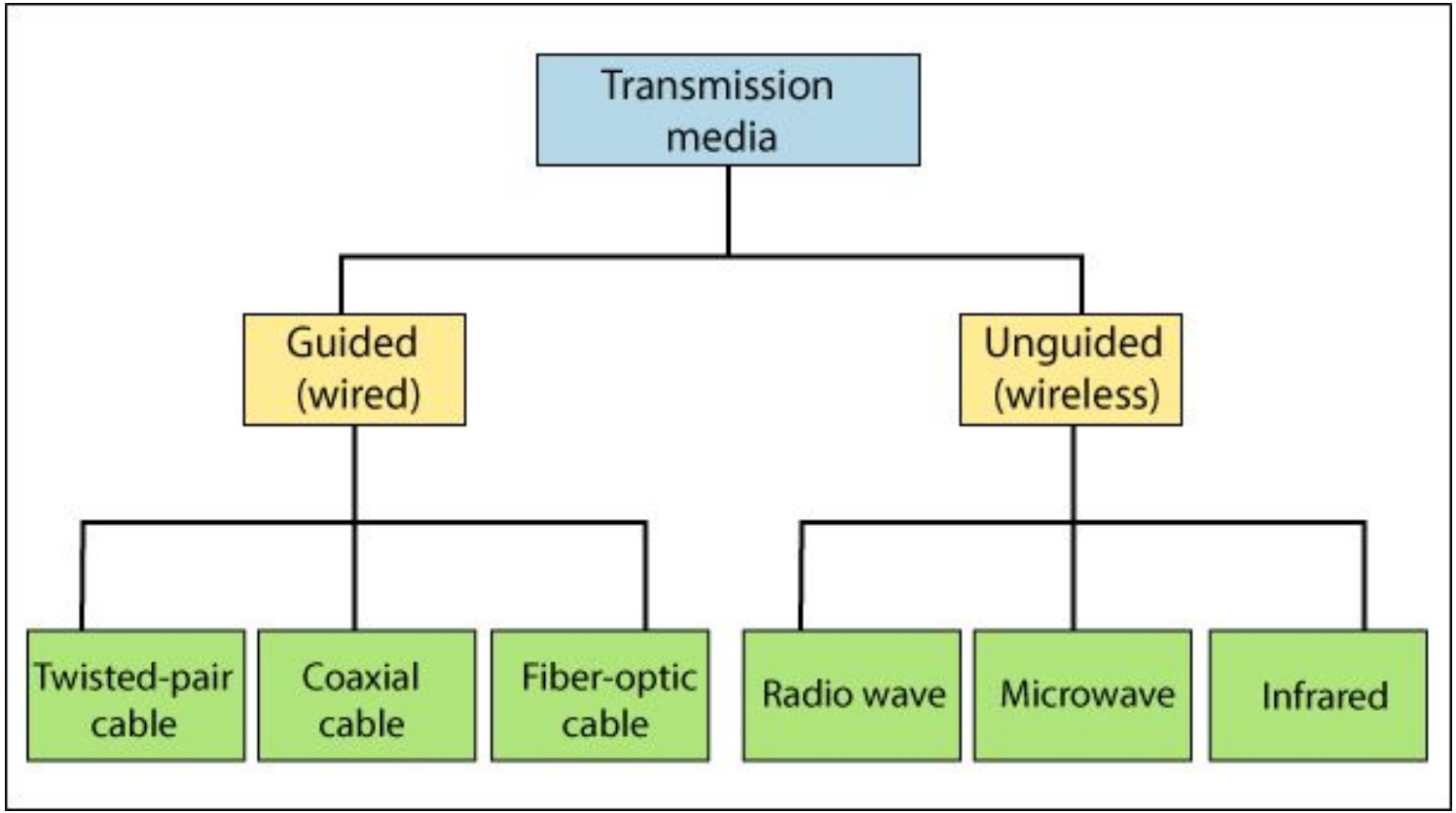
- **Transmission media** is a pathway that carries the information from sender to receiver.
- We use different types of cables or waves to transmit data.
- Data is transmitted normally through electrical or electromagnetic signals.

Description

- Transmission media are located below the physical layer
- Computers use signals to represent data.
- Signals are transmitted in form of electromagnetic energy.



Types of Transmission Media

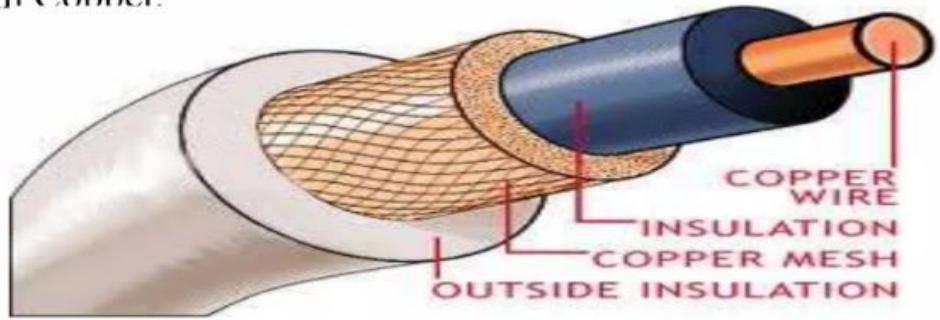


Guided Media

- Guided media are the physical links through which signals are confined to narrow path. These are also called guide media. Bounded media are made up o a external conductor (Usually Copper) bounded by jacket material. Bounded media are great for LABS because they offer high speed, good security and low cast. However, some time they cannot be used due distance communication. Three common types of bounded media are used of the data transmission. These are
- Coaxial Cable
- Twisted Pairs Cable
- Fiber Optics Cable

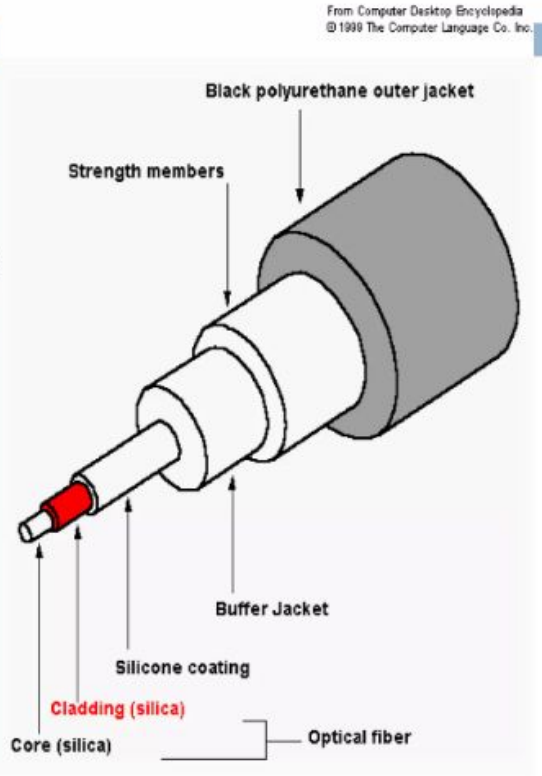
Coaxial cable

- Coaxial cable is very common & widely used commutation media. For example TV wire is usually coaxial. Coaxial cable gets its name because it contains two conductors that are parallel to each other. The center conductor in the cable is usually copper. The copper can be either a solid wire or stranded martial.
- Outside this central Conductor is a non-conductive material. It is usually white, plastic material used to separate the inner Conductor form the outer Conductor. The other Conductor is a fine mesh made from Copper.



Fiber optic cable

- Fiber optic cable uses electrical signals to transmit data. It uses light. In fiber optic cable light only moves in one direction for two way communication to take place a second connection must be made between the two devices. It is actually two stands of cable. Each stand is responsible for one direction of communication. A laser at one device sends pulse of light through this cable to other device. These pulses translated into "**1's**" and "**0's**" at the other end.
- In the center of fiber cable is a glass stand or core. The light from the laser moves through this glass to the other device around the internal core is a reflective material known as**CLADDING**. No light escapes the glass core because of this reflective**cladding.**

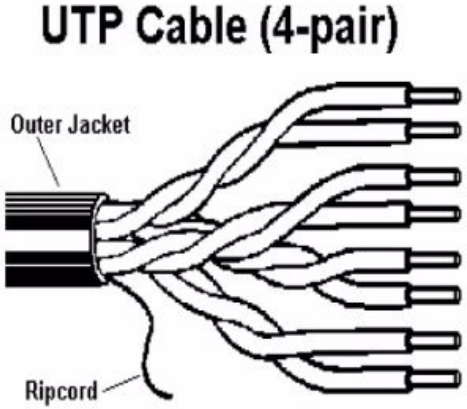


Twisted Pair Cable

- The most popular network cabling is Twisted pair. It is light weight, easy to install, inexpensive and support many different types of network. It also supports the speed of **100 mps.**Twisted pair cabling is made of pairs of solid or stranded copper twisted along each other. The twists are done to reduce vulnerably to EMI and cross talk. The number of pairs in the cable depends on the type. The copper core is usually **22-AWG or 24-AWG, as**measured on the American wire gauge standard. There are two types of twisted pairs cabling
- **1. Unshielded twisted pair (UTP)**
- **2. Shielded twisted pair (STP)**

Unshielded twisted pair (UTP)

UTP is more common. It can be either voice grade or data grade depending on the condition. UTP cable normally has an impedance of 100 ohm. UTP cost less than STP and easily available due to its many use. There are five levels of data cabling



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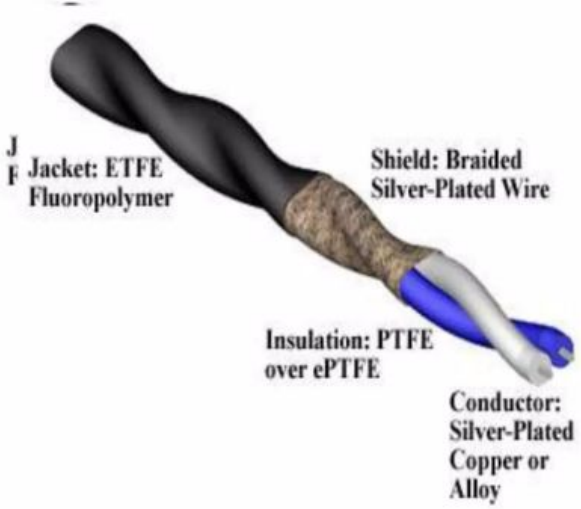
Shielded twisted pair (STP)

- It is similar to UTP but has a mesh shielding that's protects it from EMI which allows for higher transmission rate.



Unguided Media

Unguided media or wireless media doesn't use any physical connectors between the two devices communicating. Usually the transmission is send through the atmosphere but sometime it can be just across the rule. Wireless media is used when a physical obstruction or distance blocks are used with normal cable media. The three types of wireless media are:

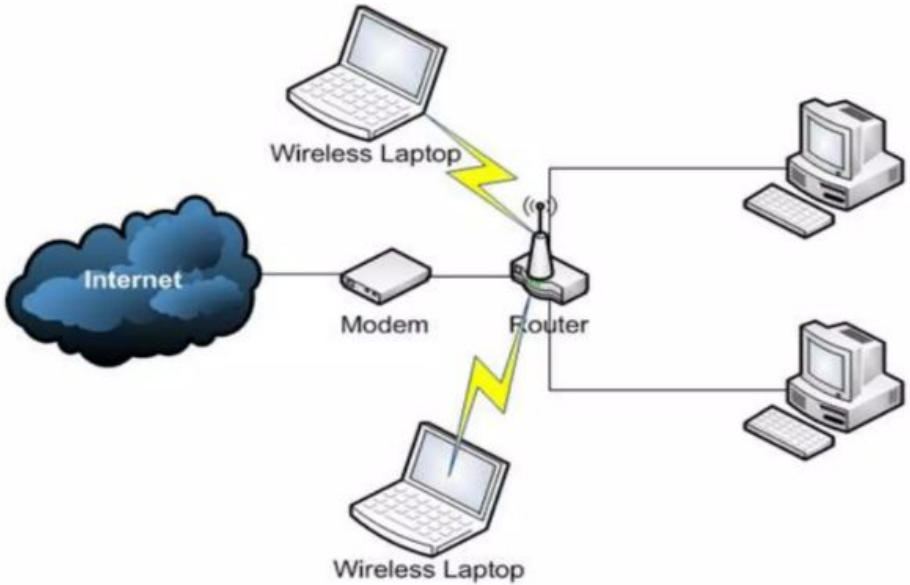
- Radio waves
- Micro waves
- Infrared waves

Devices used in networking

Mainly four devices are used in networking:-

- Modem
- Hub
- Switch
- Router

MODEM-(**mo**dulator-**dem**odulator)



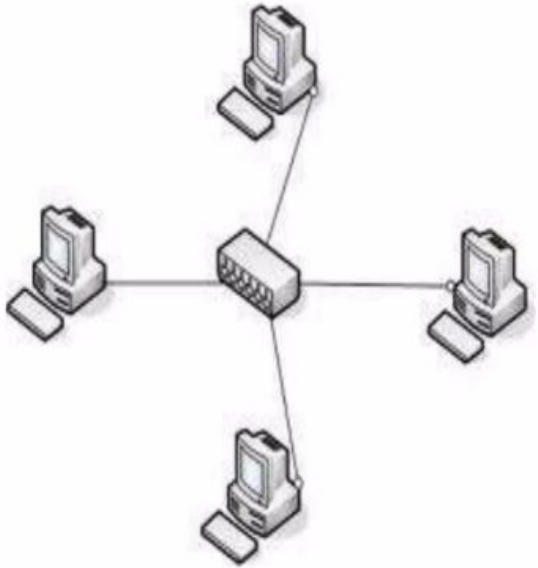
Modem

A **modem** is a device that modulates an analog carrier signal to encode digital information and also demodulates such a carrier signal to decode the transmitted information. The goal is to produce a signal that can be transmitted easily and decoded to reproduce the original digital data. Modems can be used over any means of transmitting analog signals, from light emitting diodes to radio.

There are two types of modem:-

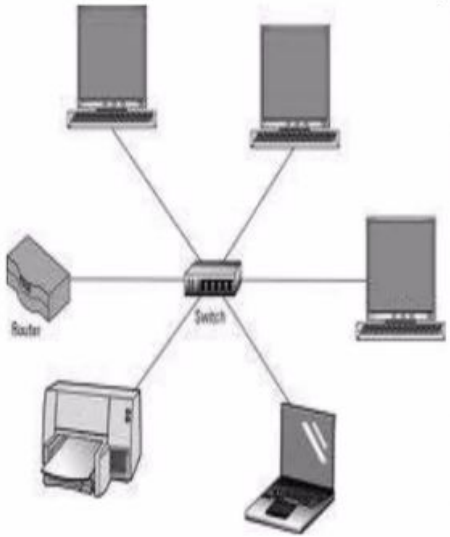
1. internal modem
2. external modem

HUB



A common connection point for devices in a network. A hub contains multiple ports. When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets.

Switch

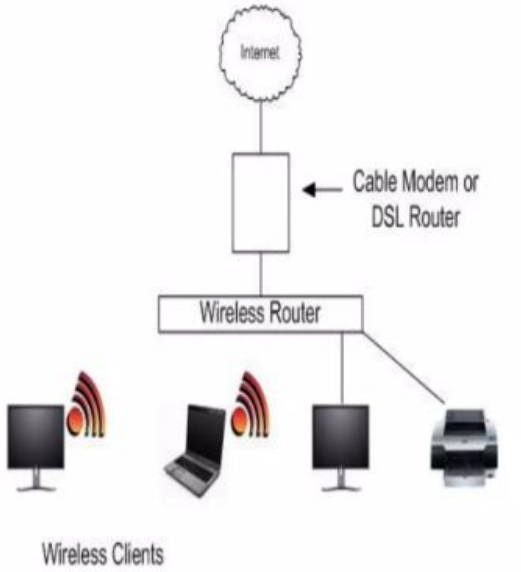


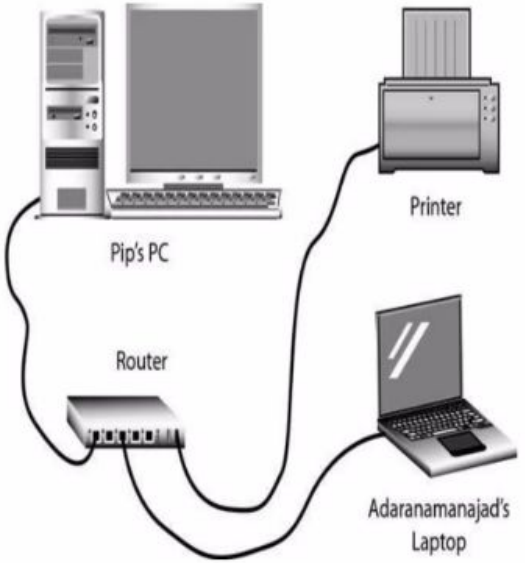
A network switch is a small hardware device that joins multiple computers together within one local area network (LAN). Network switches appear nearly identical to network hubs, but a switch generally contains more intelligence than a hub. Unlike hubs, network switches are capable of inspecting data packets as they are received, determining the source and destination device of each packet, and forwarding them appropriately. By delivering messages only to the connected device intended, a network switch conserves network bandwidth and offers generally better performance than a hub.

Router

A **router** is a device that forwards data packets between computer networks, creating an overlay internetwork. A router is connected to two or more data lines from different networks. When a data packet comes in on one of the lines, the router reads the address information in the packet to determine its ultimate destination. Then, using information in its routing table it directs the packet to the next network on its journey. Routers perform the "traffic directing" functions on the Internet. A data packet is typically forwarded from one router to another through the networks that constitute the internetwork until it gets to its destination node.

Router


Internet
Cable Modem or
DSL Router
Wireless Router
Wireless Clients


Pip's PC
Printer
Router
Adaranamanajad's
Laptop

Multiplexing

What is it?

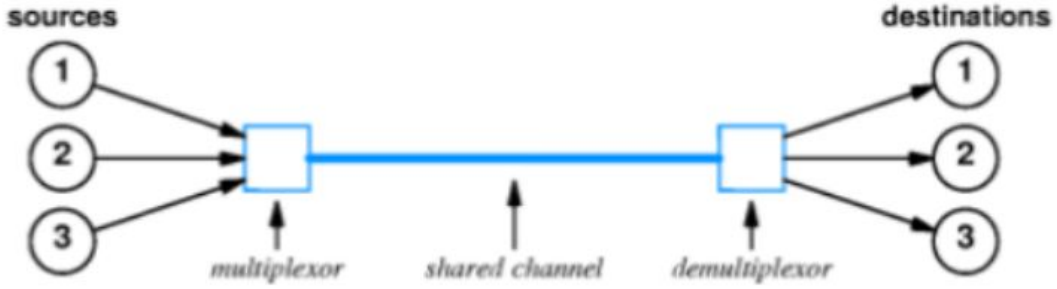
- Its a method by which multiple analog or digital signals are combined into one signal over a shared medium.
- A technique through which low-speed signals are converted into high-speed signals.
- Many to one*.
- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

Why Multiplexing

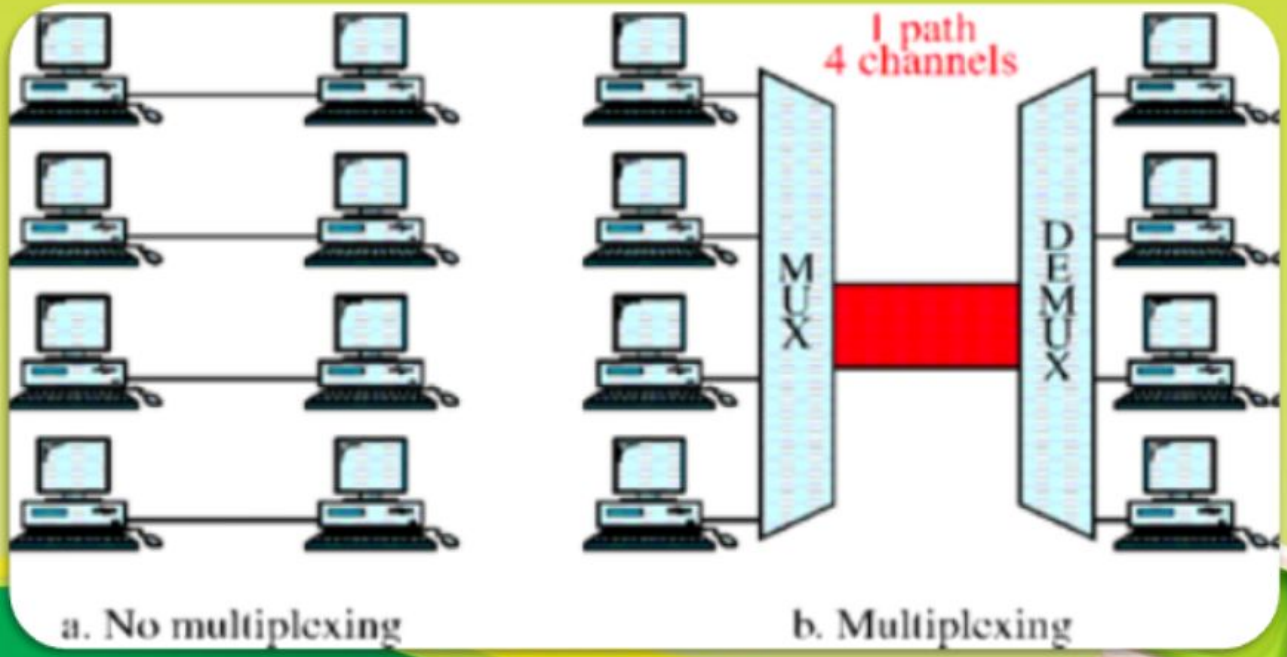
- Media Sharing
  - Medium Transmission Capacity
  - High bandwidth media (coax cable, optical fiber)
- Cost-effective
  - Medium Transmission Capacity > data rate required

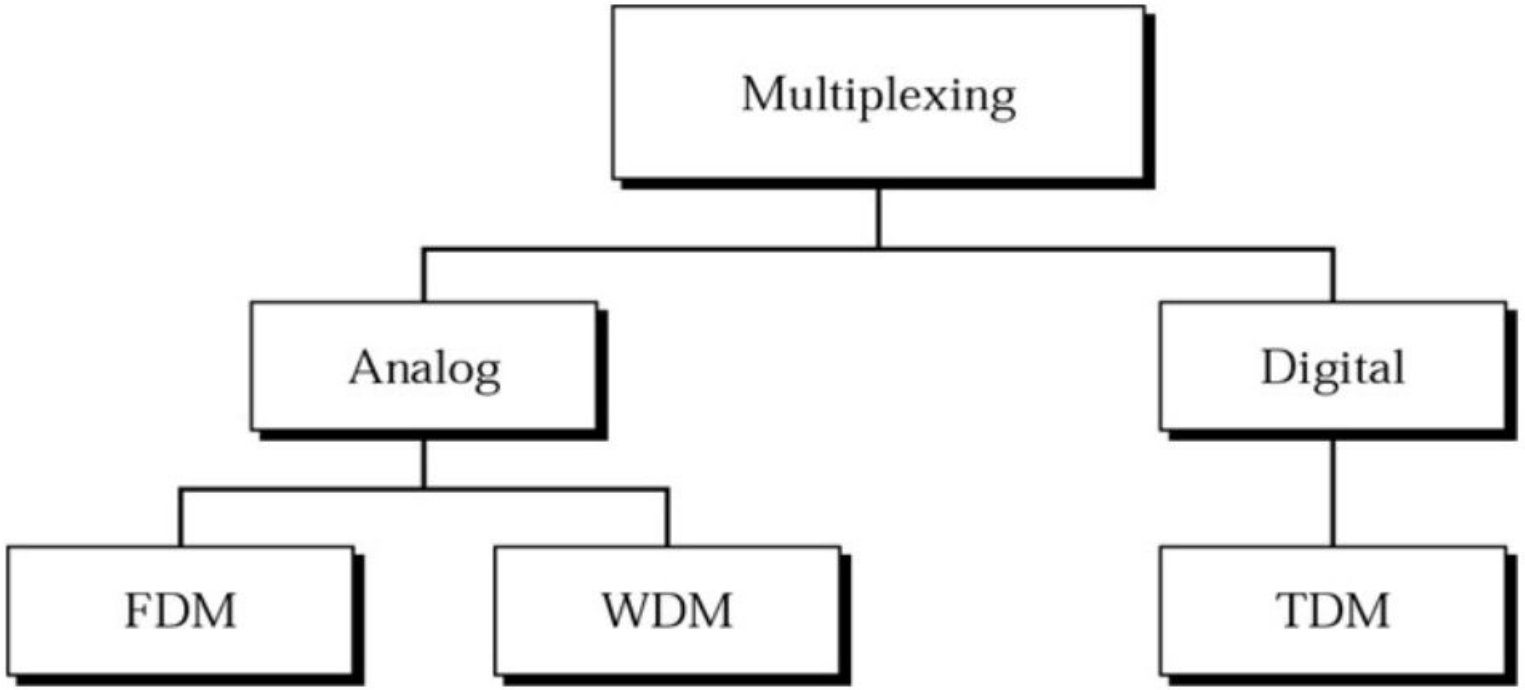
Terminology

- ***Multiplexer/MUX*** is the device which does all this function (many to one) i.e., at source end.
- ***Demultiplexer/DEMUX*** is a device which performs functions reverse to that of MUX (one to many) i.e., at destination end.



Multiplexing vs. Non-Multiplexing





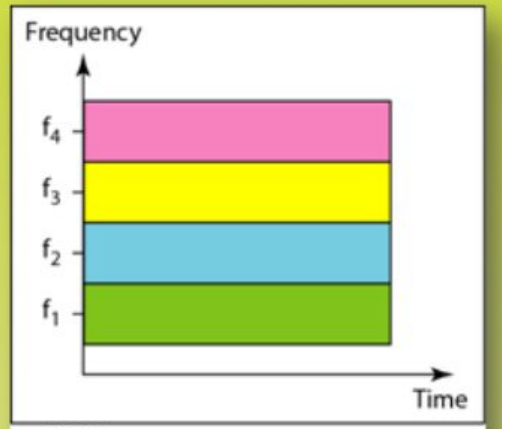
Frequency Division Multiplexing

Wavelength Division Multiplexing

Time Division Multiplexing

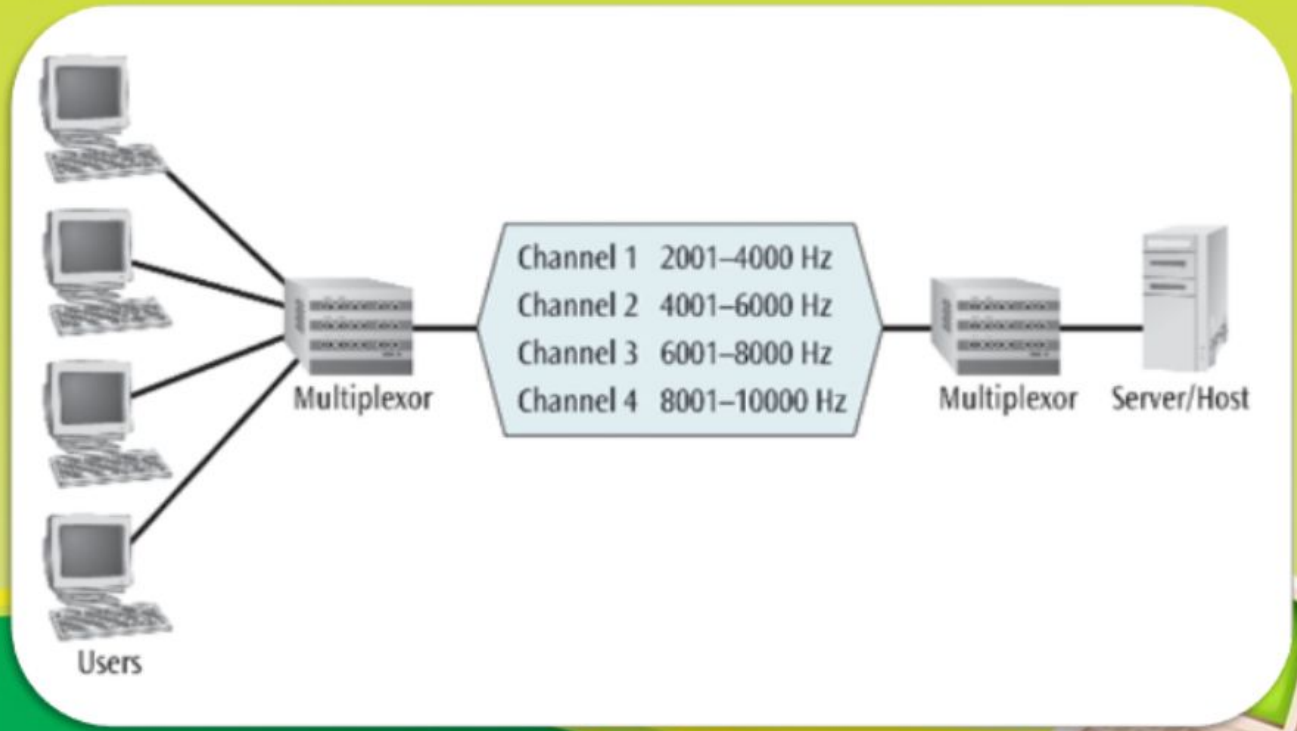
Frequency Division Multiplexing

- Total *bandwidth* available is divided into a series of non-overlapping frequency sub-bands, each of which carries a separate signal.
- All signals are transmitted at the same time, each using different frequencies.
- Bandwidth = data transfer rate i.e. Kbps, Mbps etc.
- Physical medium = Coaxial cable, fiber optic



a. FDM

Frequency Division Multiplexing

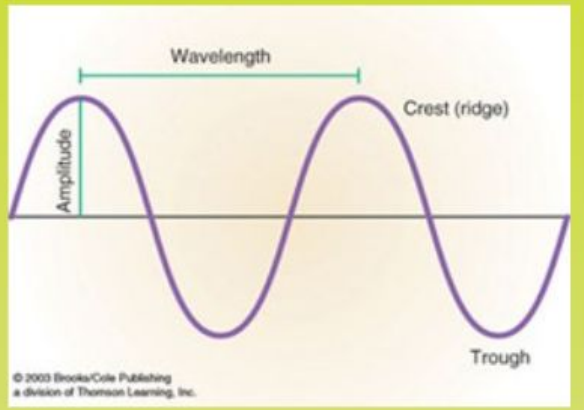


Frequency Division Multiplexing

- Analog signaling is used to transmit signals.
- Broadcast radio and television, cable television, and AMPS cellular phone systems use frequency division multiplexing.
- Oldest multiplexing technique.
- Involves analog signaling → more susceptible to noise.

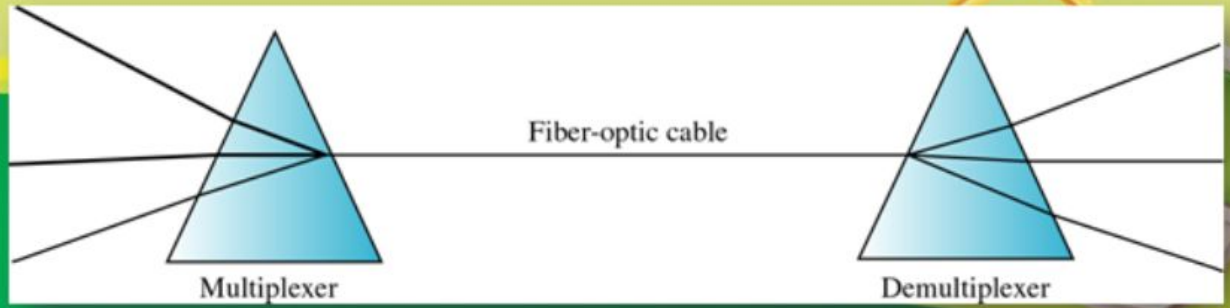
Wavelength Division Multiplexing

- ***Wavelength*** is the distance (measured in meter) b/w consecutive corresponding points of same phase, such as crests, troughs or zero crossings & is denoted by λ (read as lambda).
- In simple English, it's the distance over which the wave's shape changes.
- ***Frequency*** is the no. of cycles per second & is denoted by *f*.
- *λ* & *f* are inversely proportional to each other.

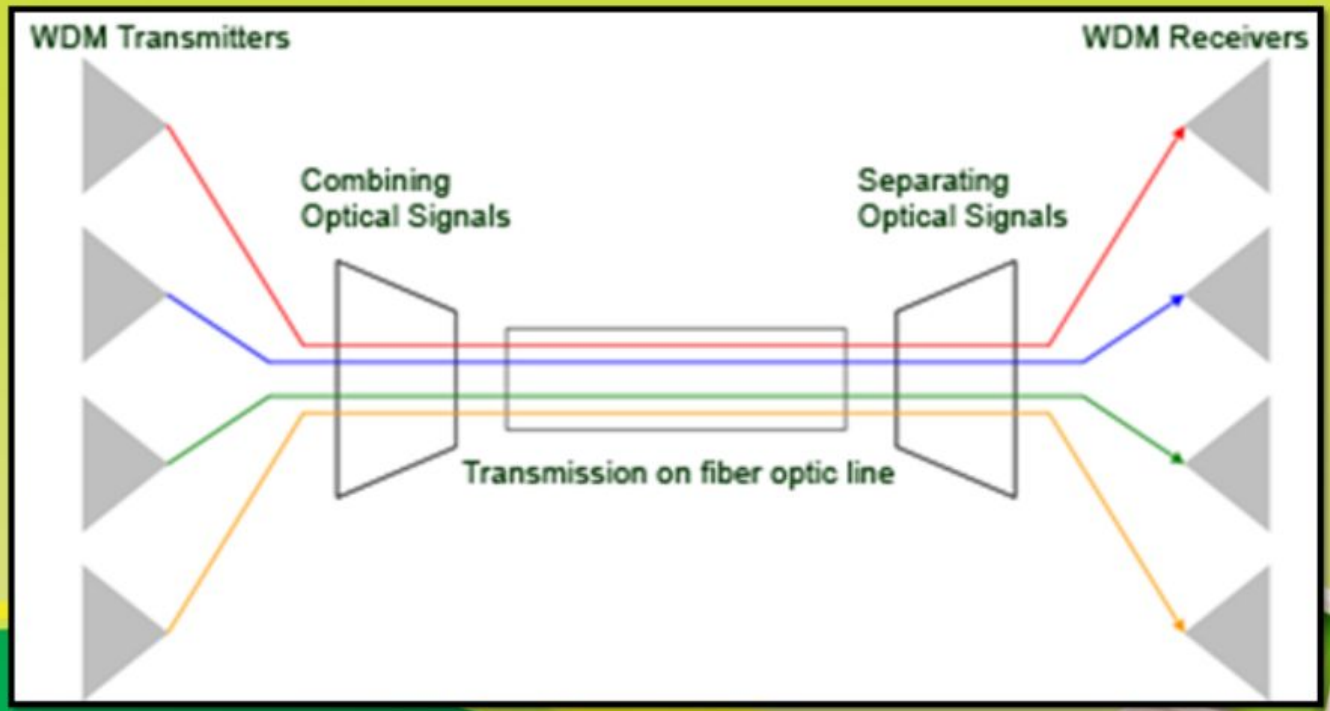


Wavelength Division Multiplexing

- A method of combining multiple signals on laser beams at various infrared (IR) wavelengths for transmission along fiber optic media.
- WDM is similar to FDM but instead of taking place at radio frequencies (RF), WDM is done in the IR portion of the electromagnetic (EM) spectrum.
- Physical medium = Fiber optics.
- Different wavelengths (i.e. colors) of laser light are muxed onto a single optical fiber.

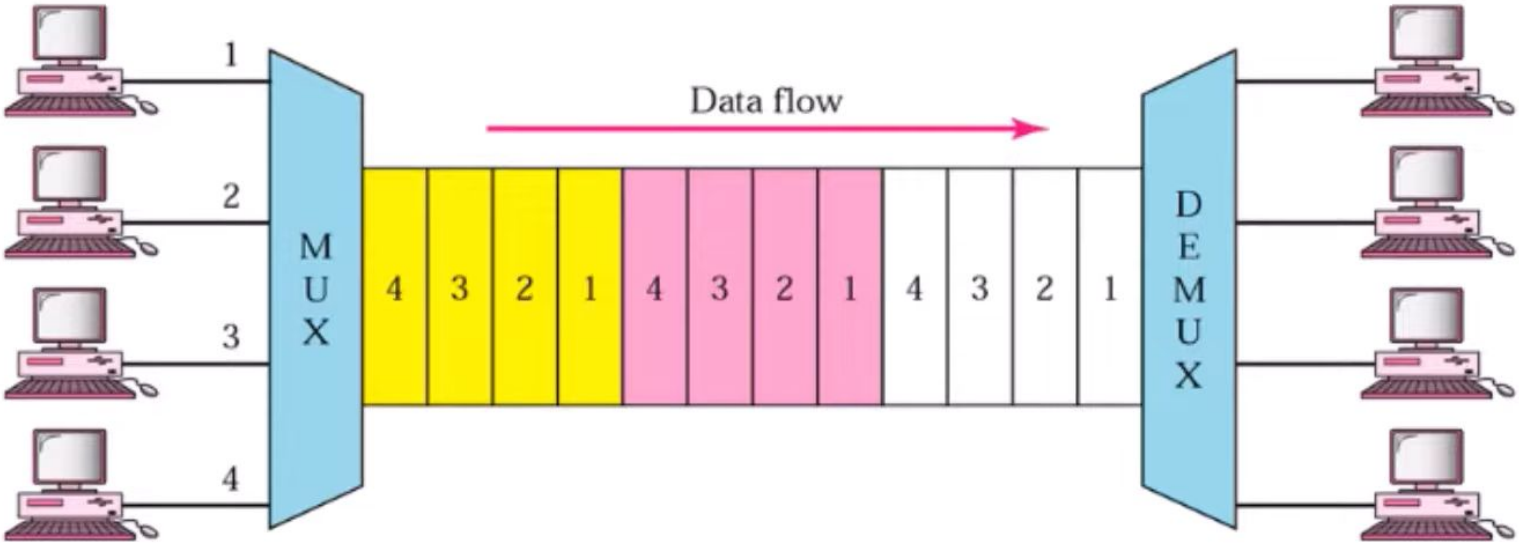


Wavelength Division Multiplexing



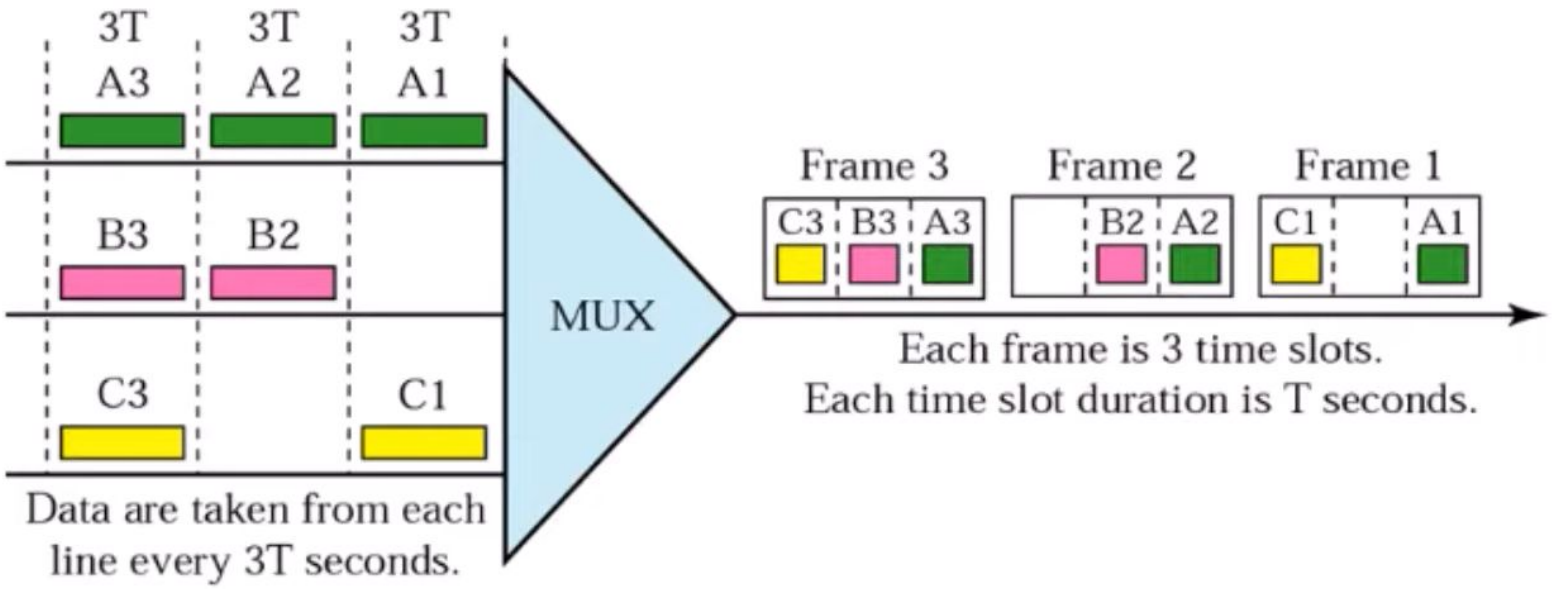
Time Division Multiplexing

- TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.
- Sharing signal is accomplished by dividing available transmission time into time segments on a medium among users.
- Instead of sharing a portion of the bandwidth as in FDM, time is shared.
- Digital signaling is used exclusively.
- TDM comes in two basic forms:
  - *Synchronous time division multiplexing (STDM)*
  - *Statistical, or asynchronous time division multiplexing (ATDM)*


1
2
3
4
MUX
Data flow
4 3 2 1 4 3 2 1 4 3 2 1
DEMUX

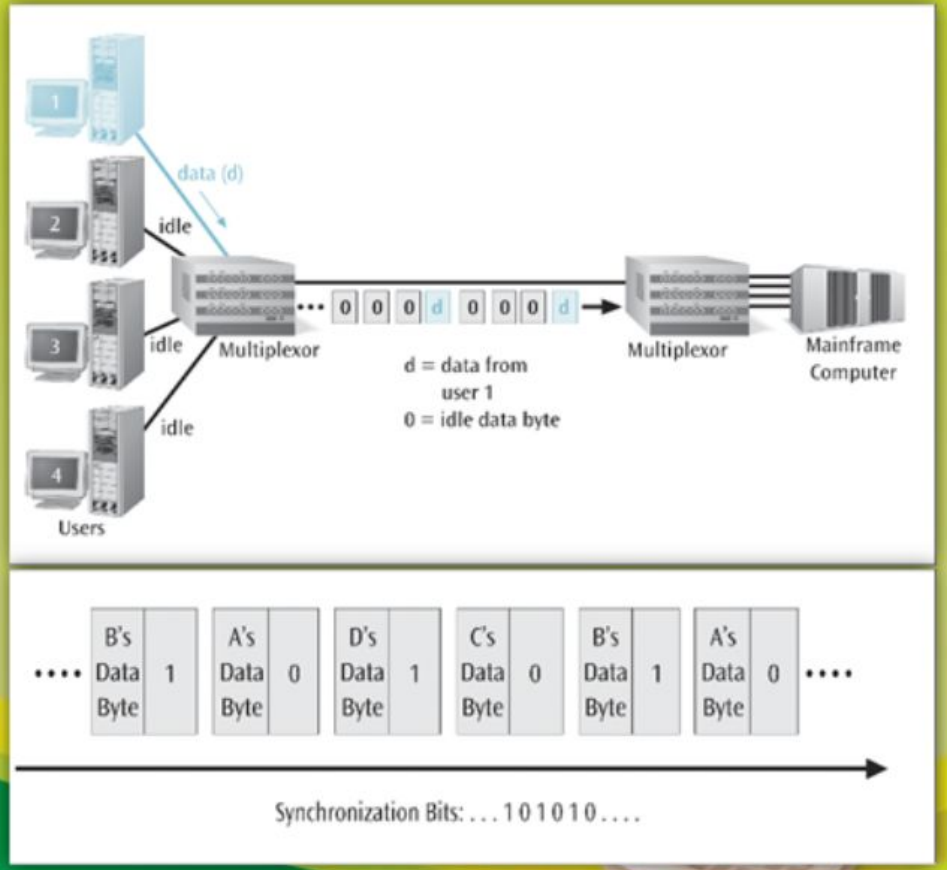
Synchronous TDM

- The original TDM.
- Multiplexor :-
  - *Accepts input from attached devices in a round-robin fashion.*
  - *Transmits data in a never ending pattern.*
- The multiplexer allocates exactly the *same timeslot* to each device at all times, whether or not a device has anything to transmit.


3T
3T
3T
A3
A2
A1
B3
B2
C3
C1
MUX
Frame 3
Frame 2
Frame 1
C3
B3
A3
B2
A2
C1
A1
Each frame is 3 time slots.
Each time slot duration is T seconds.
Data are taken from each
line every 3T seconds.

Synchronous TDM

- To keep the receiver synchronized with the incoming data stream, the transmitting multiplexor can insert alternating 1s and 0s into the data stream.



Synchronous TDM

Frame Synchronization

- Frame synchronization is needed at the TDM receiver so that the received data can be sorted and directed to appropriate output channel.
- Frame sync is provided to the receiver in two different ways:
  - Provided to the demux by sending a frame sync signal from the transmitter over a separate channel.
  - Derive the frame sync from the TDM signal itself.

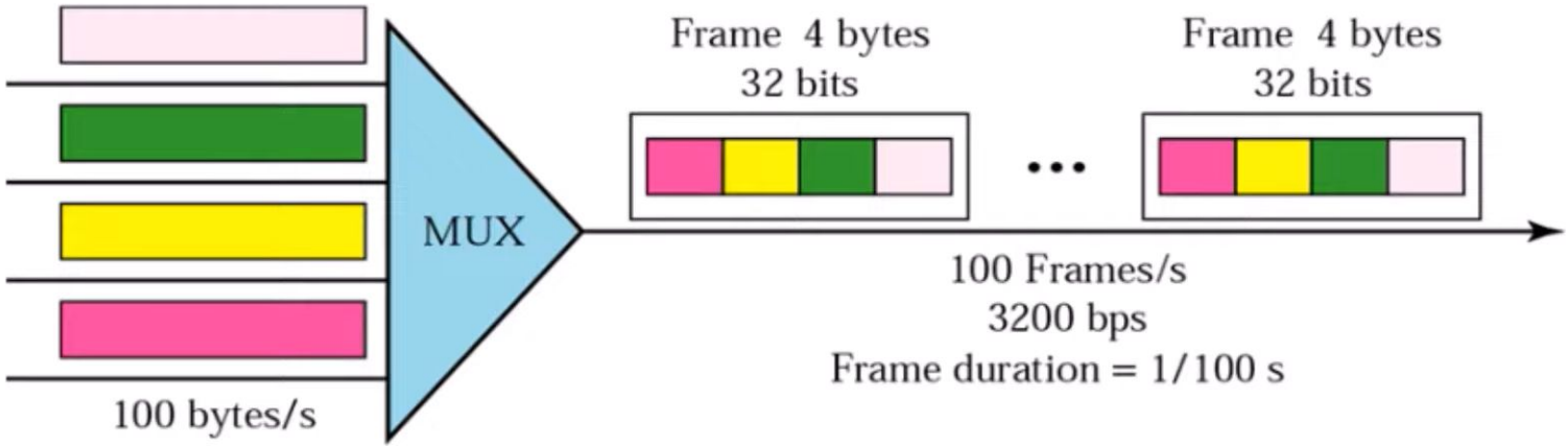
Synchronous TDM

Frame Synchronization

- Frame synchronization is needed at the TDM receiver so that the received multiplexed data can be sorted and directed to appropriate output channel.
- Frame sync is provided to the receiver in two different ways:
  - provided to the de-multiplexer circuit by sending a frame sync signal from the transmitter over a separate channel.
  - derive the frame sync from the TDM signal itself.

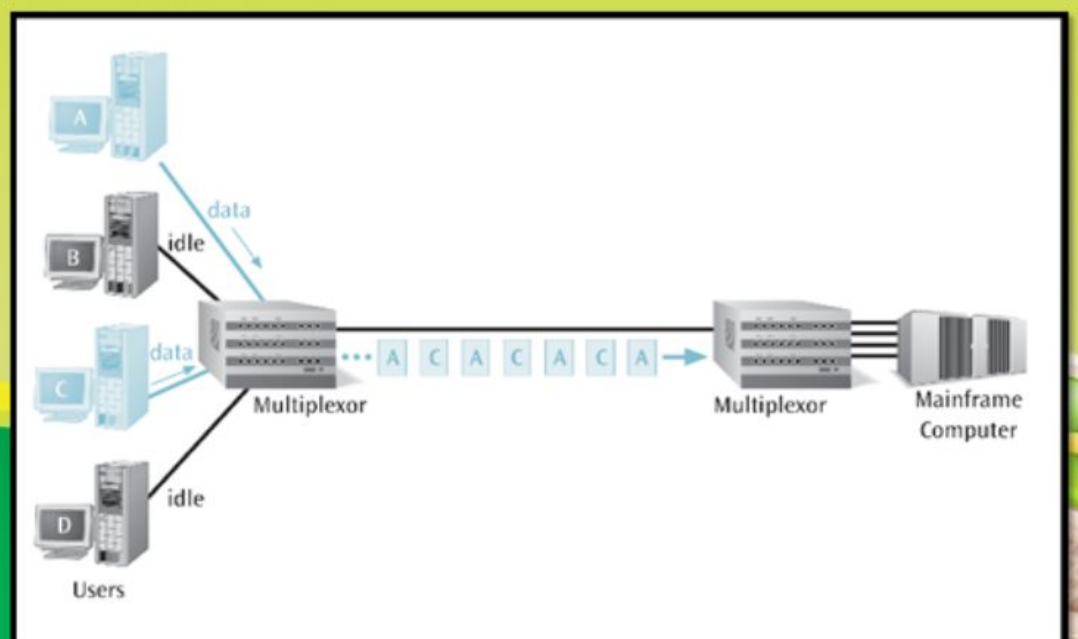
Asynchronous TDM

- Statistical multiplexor - transmits only the data from active workstations.
  - If a workstation is not active, no space is wasted on the multiplexed stream.
- A statistical multiplexor
  - Accepts incoming data streams.
  - Creates a frame containing only the data to be transmitted.


Frame 4 bytes
32 bits
Frame 4 bytes
32 bits
MUX
100 Frames/s
3200 bps
Frame duration = 1/100 s
100 bytes/s

Asynchronous TDM

- Timeslots are allocated as needed dynamically rather than pre-assigned to specific transmitters.
- ATDM is more intelligent and has better bandwidth efficiency than TDM.



Asynchronous TDM

- STDM is often used for managing data being transmitted via a local area network (LAN) or a wide area network (WAN).
- An STDM adds an address field to each time slot in the frame and does not transmit empty frames.
- STDM uses dynamic time slot lengths that are variable.
- Communicating devices that are very active will be assigned greater timeslot lengths than devices that are less active.

Asynchronous TDM

- STDMs have buffer memory for temporary data storage.
- STDM uses intelligent devices capable of identifying when a terminal is idle.
- Each STDM transmission carries channel identifier (sender's address) information.
- Which includes source device address and a count of the number of data characters that belong to the listed source address.
- Channel identifiers are extra and considered as *overhead.*

Switching

- It describes how data is forwarded across an inter network.
- Determines when and how packets/messages are forwarded through the network.
- It comes under the functionality of *'Network layer'* in 7-layer OSI model which performs Path determination & logical addressing.
- Specifies the granularity and timing of packet progress.
- There are 4 types of Switching techniques;
  - Circuit Switching
  - Packet Switching
  - Message Switching
  - Cell Switching

Circuit Switching

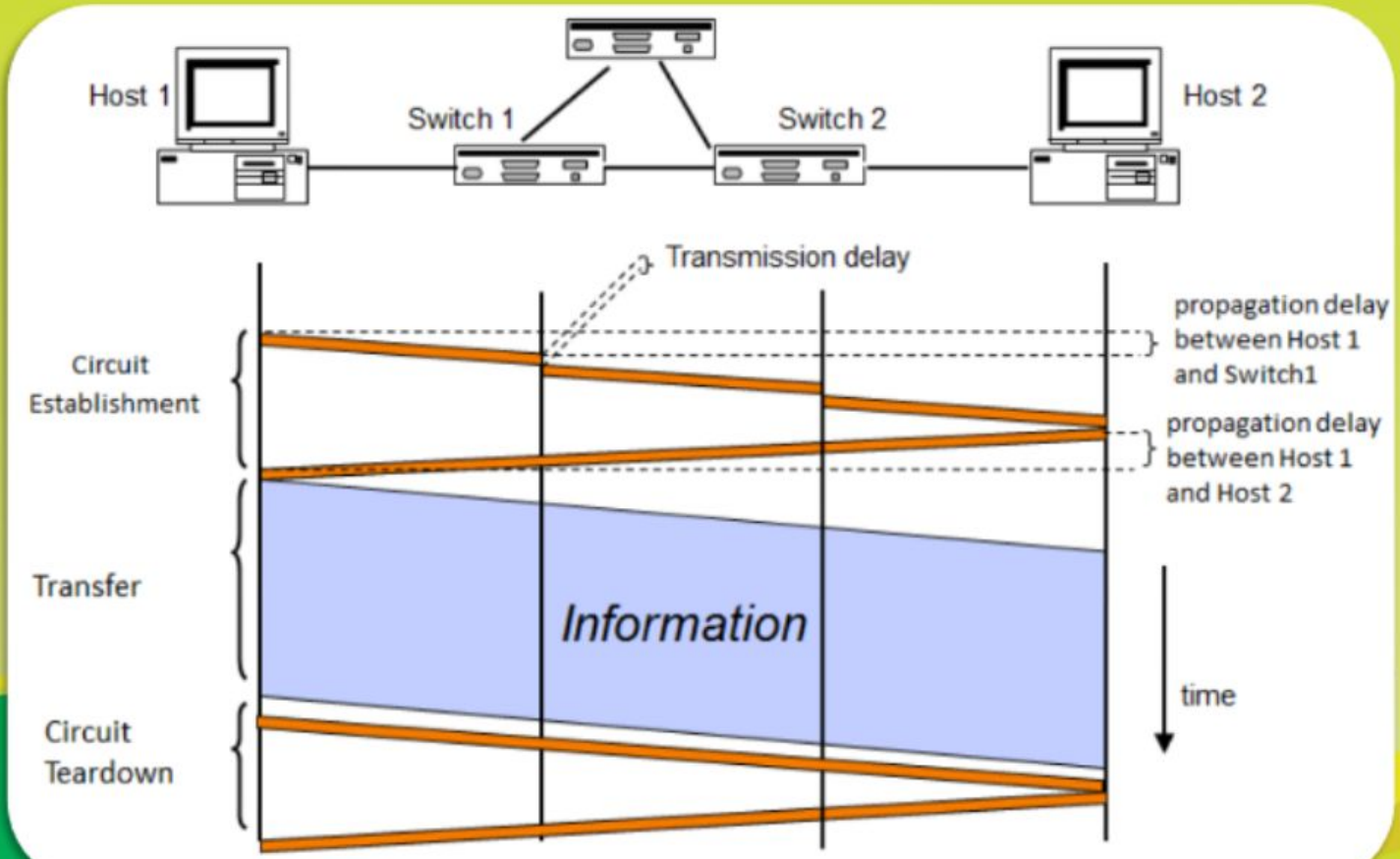
- Its a technique that directly connects the sender and the receiver in an unbroken path.
- Telephone switching equipment, for example, establishes a path that connects the caller's telephone to the receiver's telephone by making a physical connection.
- Network nodes establish a dedicated communications channel (called circuit) through the network before the nodes may communicate.
- Two phase protocols = Path Setup + Data transfer.

Circuit Switching

3 simple steps

1. **Establish**: End-to-end dedicated circuits between clients
   - Client can be a person or equipment (router or switch).
2. **Transfer**: Source sends data over the circuit
   - No destination address, since nodes know path.
3. **Teardown**: Source tears down the circuit after sending data.

Circuit Switching



Circuit Switching : Advantages

Guaranteed bandwidth

- Predictable communication performance.

Simple abstraction

- Reliable communication channel between hosts
- No worries about lost or out-of-order packets

Simple forwarding

- Forwarding based on time slot or frequency
- No need to inspect a packet header

Circuit Switching : Disadvantages

Connection Set-up delay

- No communication until the connection is set up
- Unable to avoid extra latency for small data transfers

Network state

- Network nodes must store per-connection information

Blocked connections

- Connection refused when resources are not sufficient

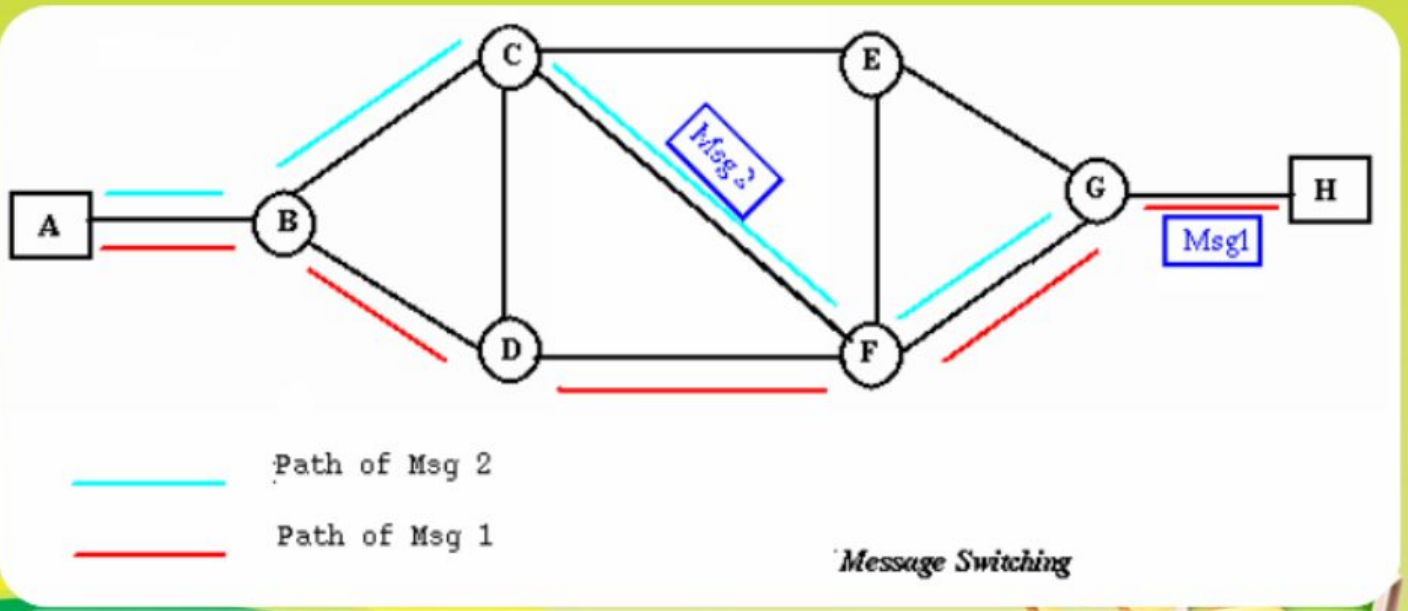
Costly

- More expensive than any other switching techniques,
- because a dedicated path is required for each connection.

Message Switching

- No need to establish a dedicated path between two stations.
- When a station sends a message, the destination address is appended to the message.
- The message is then transmitted through the network, in its entirety (in whole network), from node to node.
- Each node receives the entire message, stores it, & then transmits it to the next node.
- This type of network is called a store-and-forward network

Message Switching

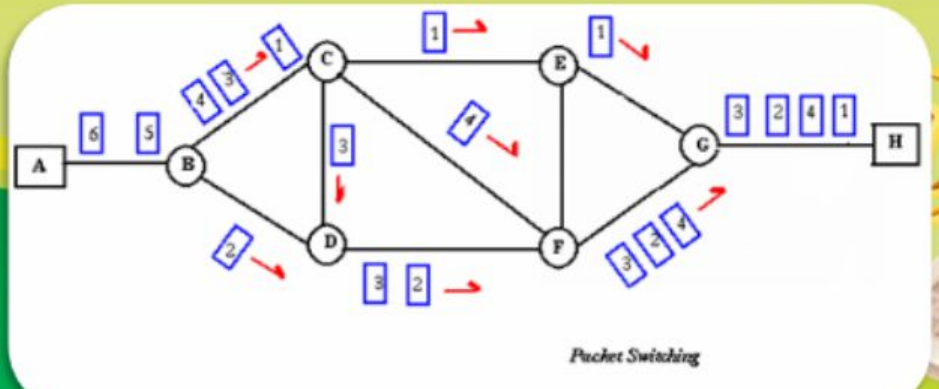


Packet Switching

- PSNs move data in separate, small blocks called packets.
- When received, packets are reassembled in the proper sequence to make up the message.
- Packet switching combines advantages of both message & circuit switching.
- There are two methods in Packet Switching:
  - Datagram.
  - Virtual Circuit (VC).

Packet Switching

- In both methods, a message is broken into small parts, called packets.
- Each packet is tagged with appropriate source and destination addresses.
- Since packets have a strictly defined maximum length, they can be stored in main memory instead of disk, therefore access delay and cost are minimized.
- Also the transmission speeds, between nodes, are optimized.

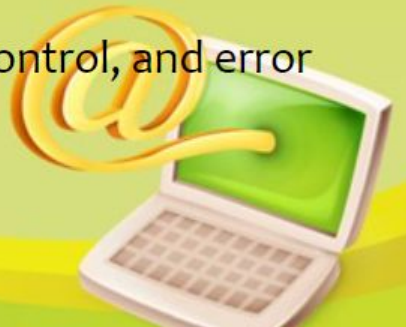


Packet Switching : Datagram

- *Definition*: *"A self-contained, independent entity of data carrying sufficient information to be routed from the source to the destination computer without reliance on earlier exchanges"*
- Similar to message switching in that each packet is a self-contained unit with complete addressing information attached.
- This fact allows packets to take a variety of possible paths through the network.
- So the packets, each with the same destination address, do not follow the same route, and they may arrive out of sequence at the exit point node (or the destination).
- Reordering is done at the destination point based on the sequence number of the packets.

Packet Switching : Virtual Circuit

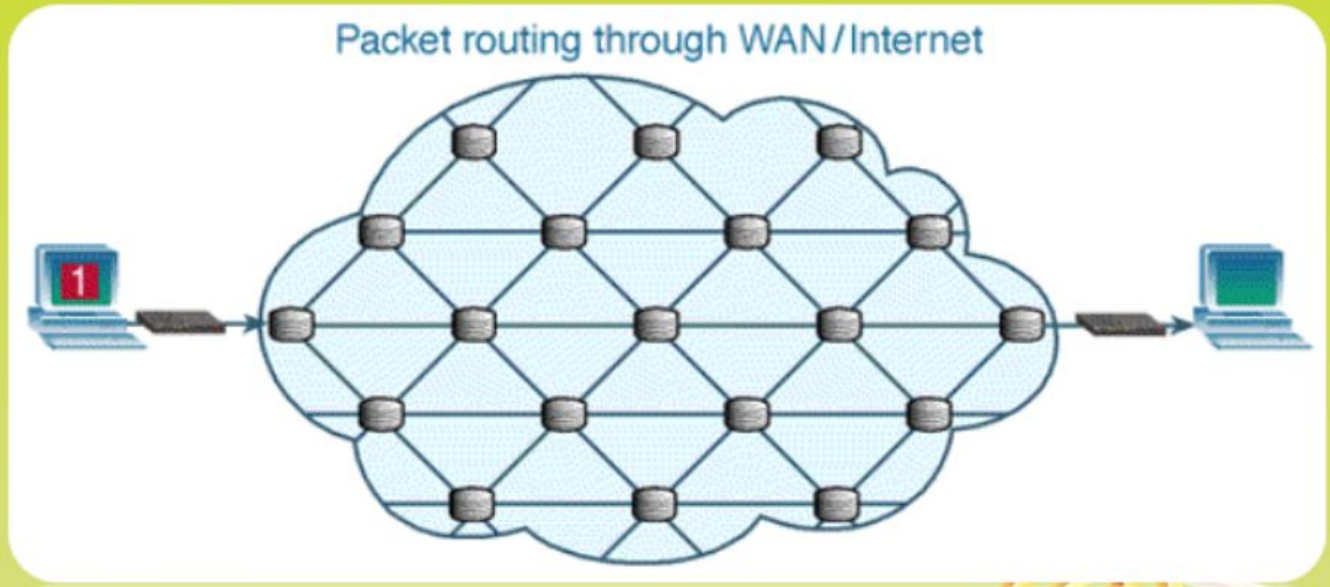
- A preplanned route is established before any data packets are sent.
- A logical connection is established when a sender sends a "call request packet" to the receiver & the receiver sends back an acknowledgement packet "call accepted packet" to the sender if the receiver agrees.
- The conversational parameters can be maximum packet sizes, path to be taken, and other variables necessary to establish and maintain the conversation.
- Virtual circuits imply acknowledgements, flow control, and error control, hence, are reliable.



Packet Switching : Virtual Circuit

- In virtual circuit, the route between stations does not mean that this is a dedicated path, as in circuit switching.
- A packet is still buffered at each node and queued for output over a line.
- Difference between virtual circuit and datagram approach:
  - With virtual circuit, the node does not need to make a routing decision for each packet.
  - It is made only once for all packets using that virtual circuit.

Packet Switching



Communication Protocols

- Describes the rules that govern the transmission of data over the communication Network.
- Provide a method for orderly and efficient exchange of data between the sender and the receiver.

Roles of Communication Protocol

1. Data Sequencing – to detect loss or duplicate packets.
2. Data Routing – to find the most efficient path between source and a destination.
3. Data formatting – defines group of bits within a packet which constitutes data, control, addressing and other information.
4. Flow control – ensures resource sharing and protection against traffic congestion by regulating the flow of data on communication lines.

Roles of Communication Protocol

5. Error control – detect errors in messages. Method for correcting errors is to retransmit the erroneous message block.
6. Precedence and order of transmission – condition all nodes about when to transmit their data and when to receive data from other nodes. Gives equal chance for all the nodes to use the communication channel.
7. Connection establishment and termination –
8. Data security – Prevents access of data by unauthorized users.

The OSI Model

- OSI is short for Open Systems Interconnection.
- OSI model was first introduced by the International Organization for Standardization (ISO) in 1984
  - Outlines **WHAT** needs to be done to send data from one computer to another.
  - Protocols stacks handle how data is prepared for transmittal
- Contains specifications in 7 different layers that interact with each other.

What is "THE MODEL?"

- Commonly referred to as the OSI reference model.
- Open system interconnection (OSI) model is a framework for defining standards for linking heterogeneous computer systems, located anywhere.
- The OSI model is a theoretical blueprint that helps us understand how data gets from one user's computer to another.
- It is also a model that helps develop standards so that all of our hardware and software talks nicely to each other.

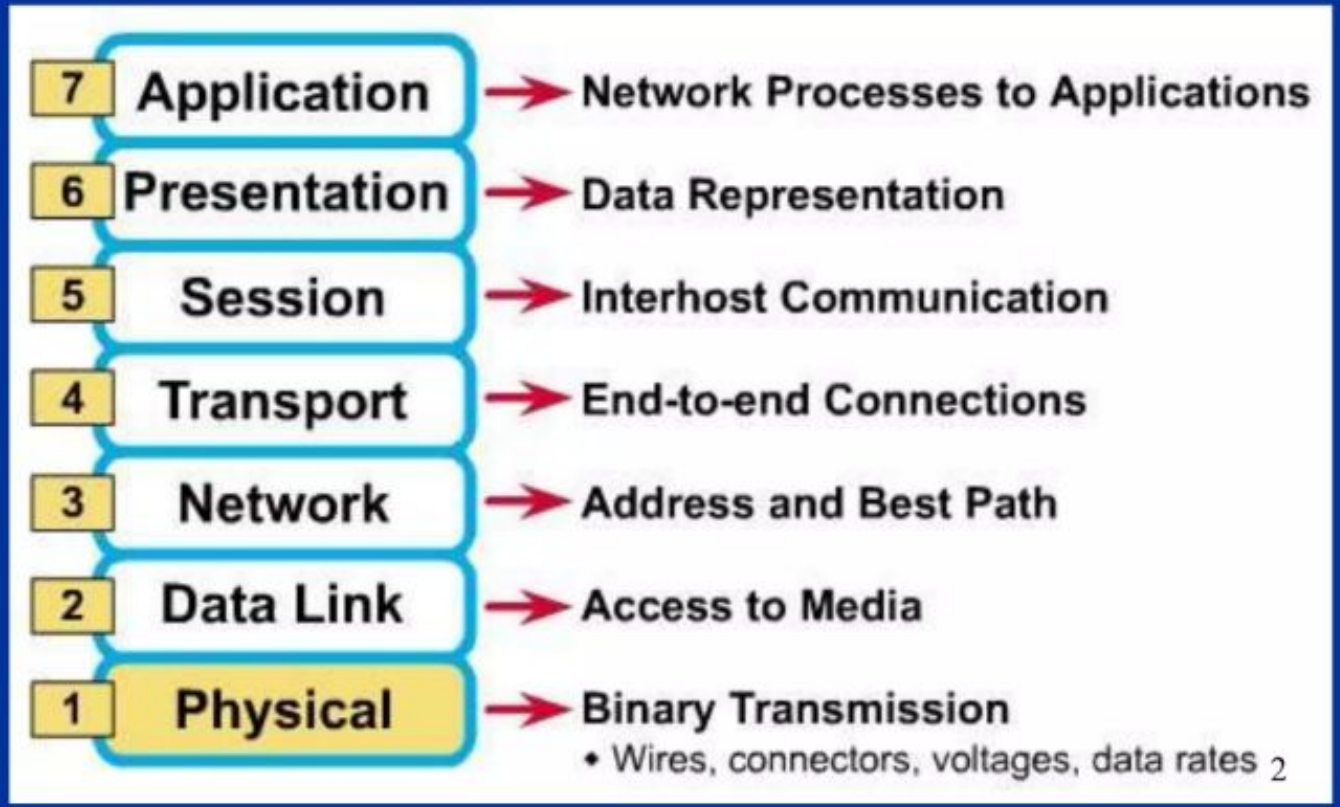
7 Layer OSI Model

- Why use a reference model?
  - Serves as an outline of rules for how protocols can be used to allow communication between computers.
  - Each layer has its own function and provides support to other layers.
- Other reference models are in use.
  - Most well known is the TCP/IP reference model.

7 Layer OSI Model

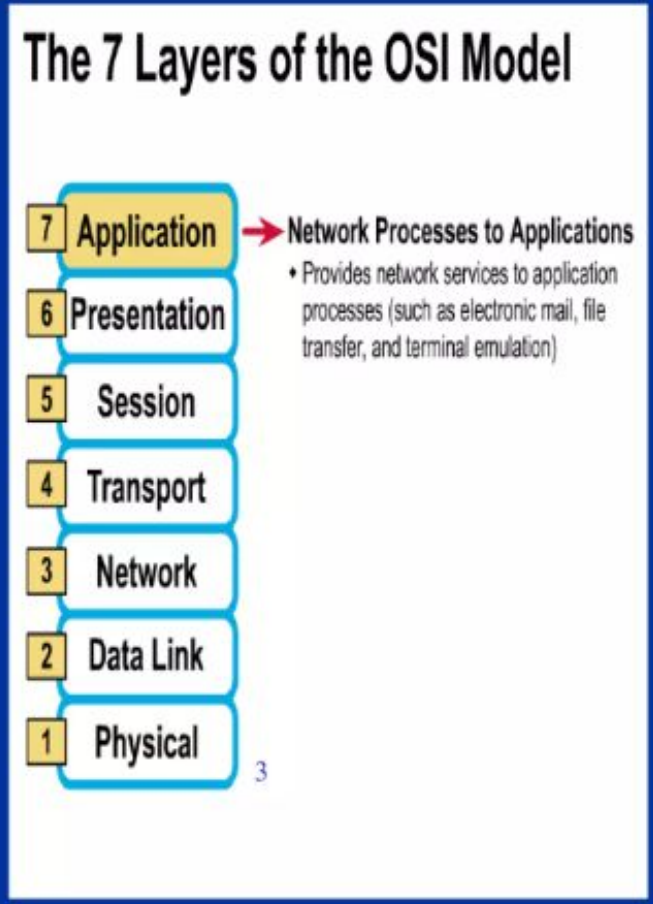
Open system interconnection (OSI) model is a framework for defining standards for linking heterogeneous computer systems, located anywhere.

What Each Layer Does



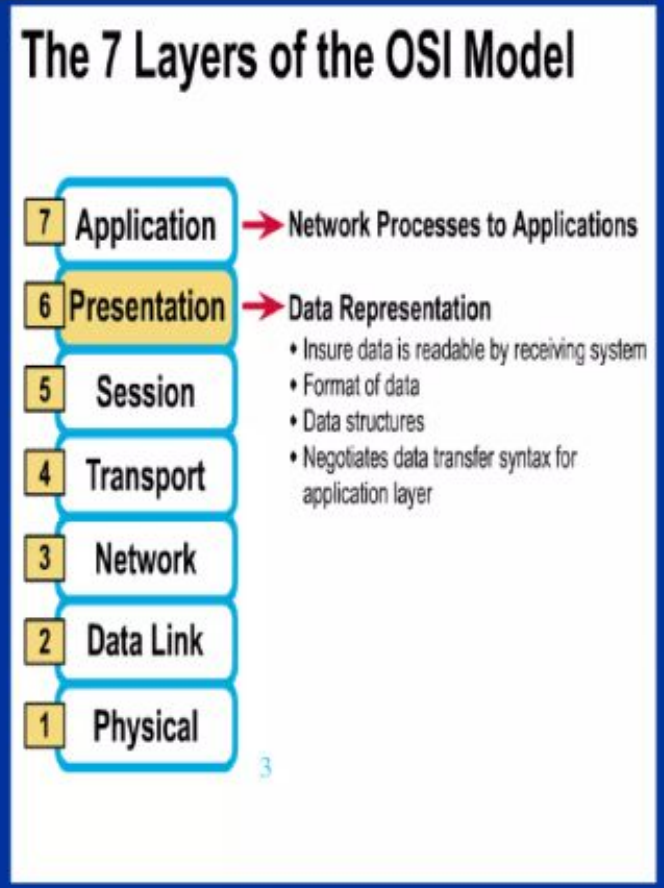
Application Layer

- Gives end-user applications access to network resources
- Where is it on my computer?
  - Workstation or Server Service in MS Windows



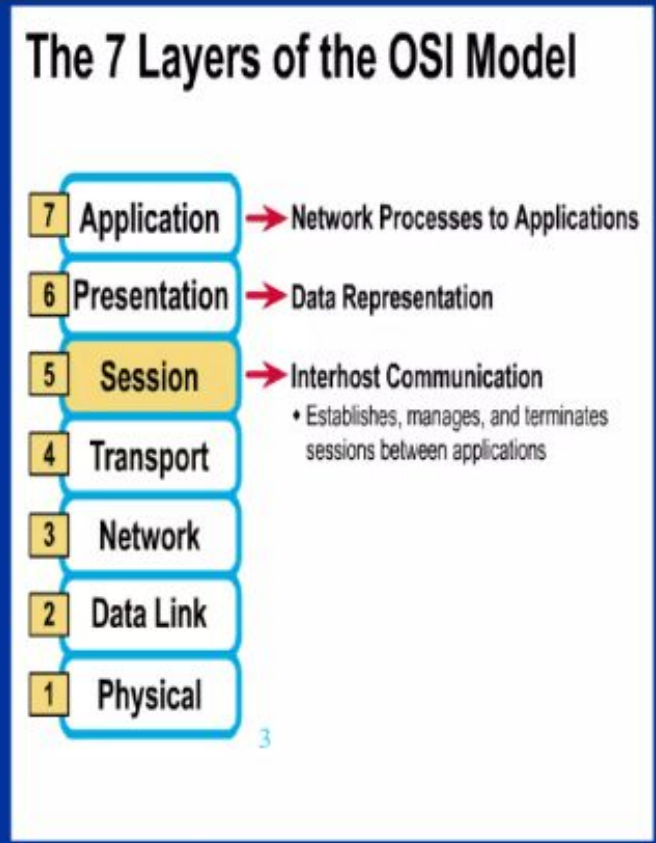
Presentation Layer

- Provides common data formatting between communicating devices
- Components make sure the receiving station can read the transferred data



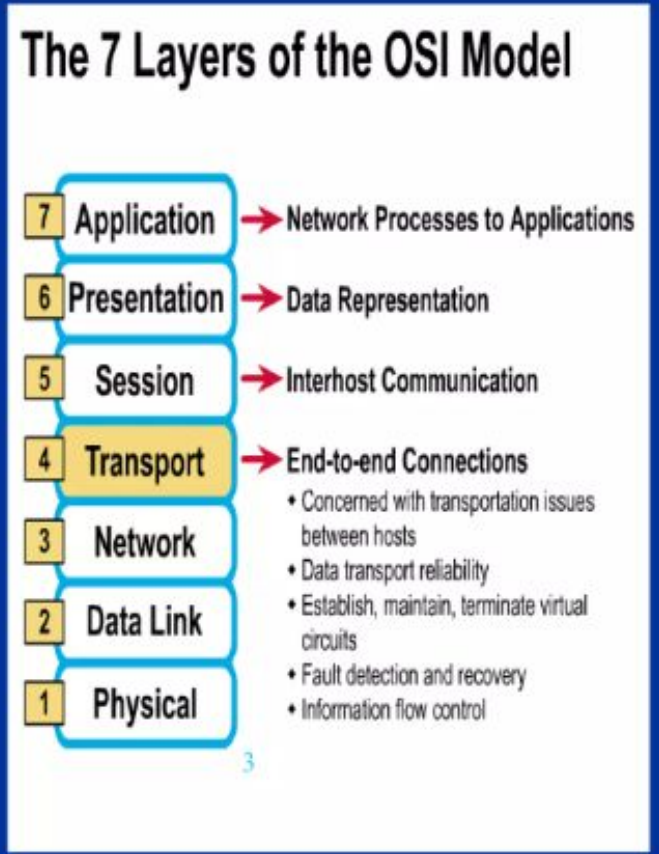
Session Layer

- Allows applications to maintain an ongoing session
- Example – NetBIOS
- Where is it on my computer?
  - Workstation and Server Service (MS)
  - Windows Client for NetWare (NetWare)



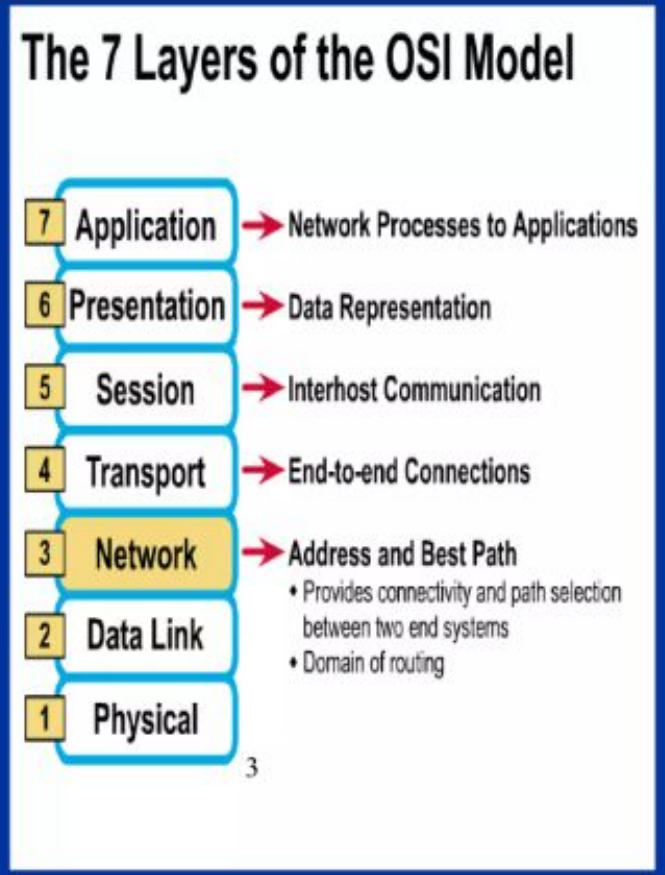
Transport Layer

- Provides reliable data delivery
- It's the TCP in TCP/IP
- Receives info from upper layers and segments it into packets
- Can provide error detection and correction



Network Layer

- Provides network-wide addressing and a mechanism to move packets between networks (routing)
- Responsibilities:
  - Network addressing
  - Routing
- Examples:
  - IP from TCP/IP
  - IPX from IPX/SPX

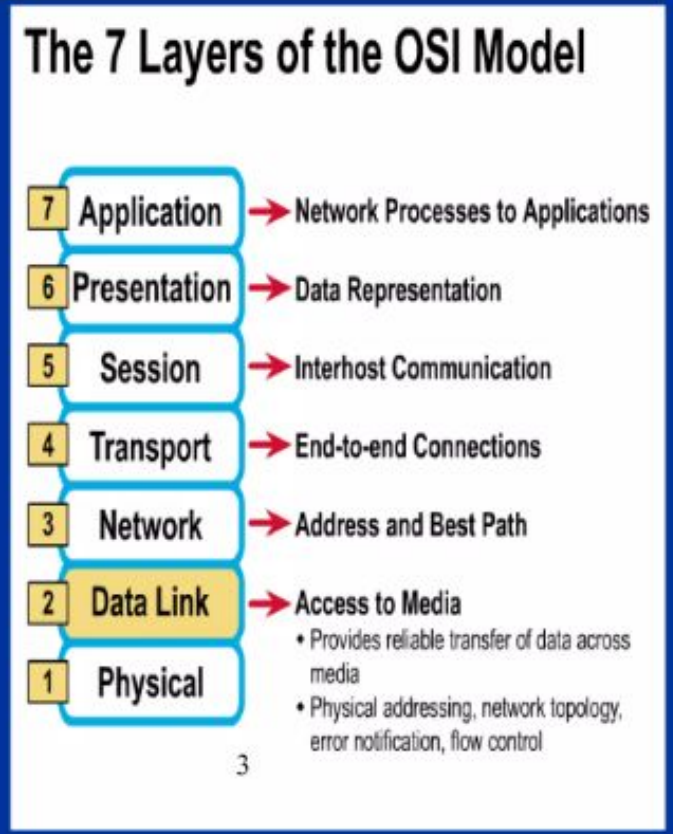


Network Addresses

- Network-wide addresses
- Used to transfer data across subnets
- Used by routers for packet forwarding
- Example:
  - IP Address
- Where is it on my computer?
  - TCP/IP Software

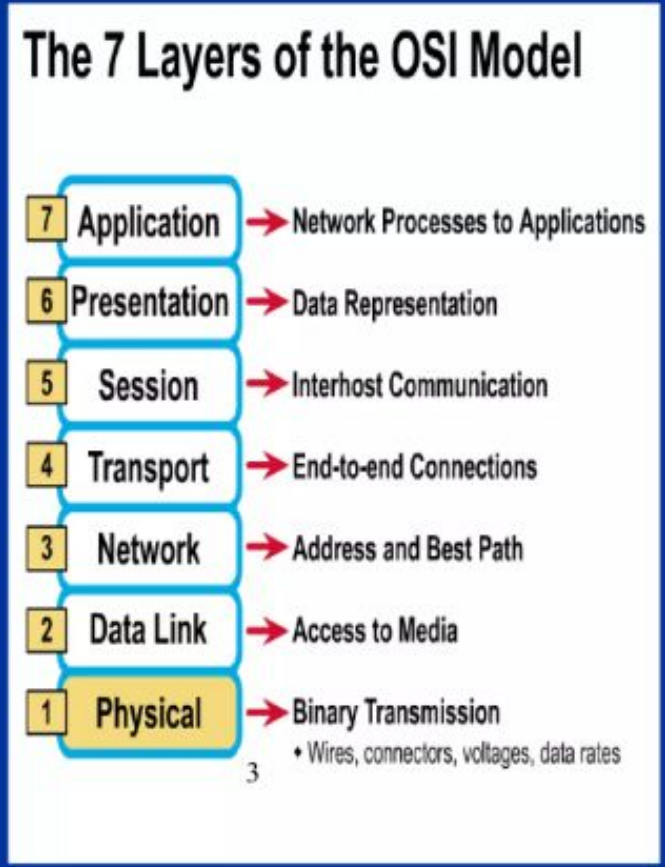
Data Link Layer

- Places data and retrieves it from the physical layer and provides error detection capabilities



Physical Layer

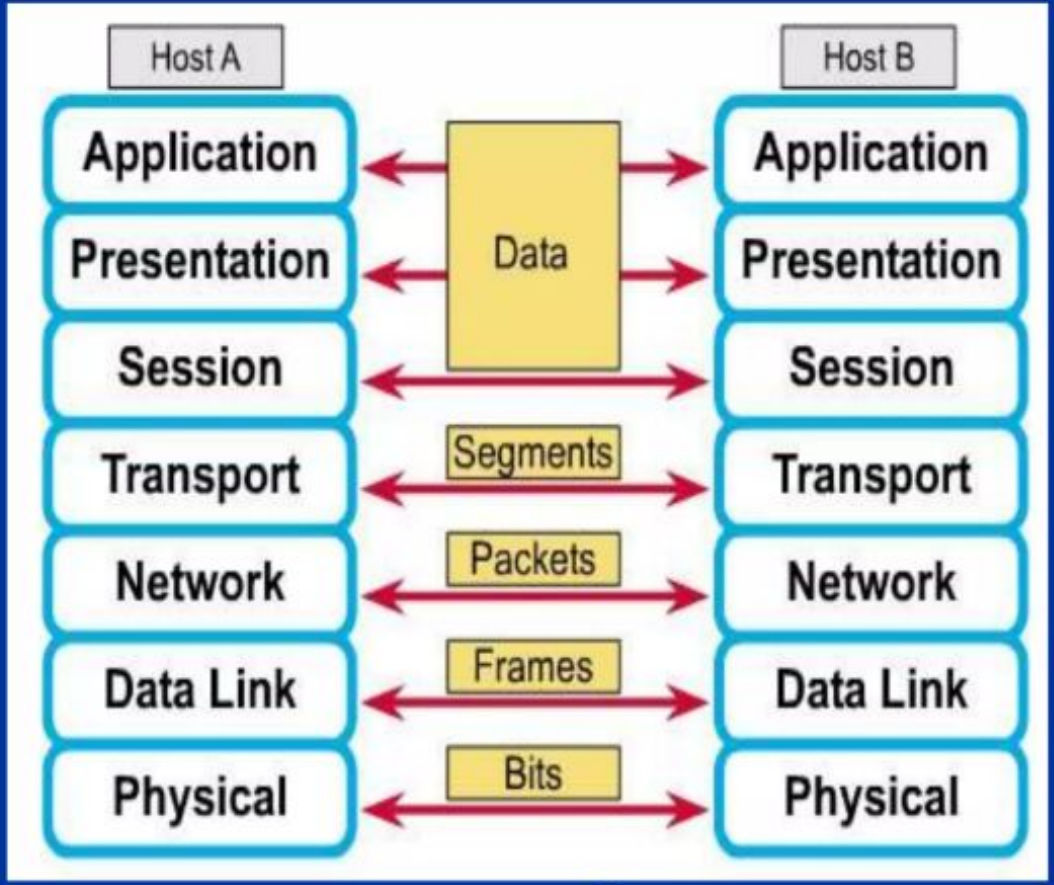
- Determines the specs for all physical components
  - Cabling
  - Interconnect methods (topology / devices)
  - Data encoding (bits to waves)
  - Electrical properties
- Examples:
  - Ethernet (IEEE 802.3)
  - Token Ring (IEEE 802.5)
  - Wireless (IEEE 802.11b)



Physical Layer (cont'd)

- What are the Physical Layer components on my computer?
- NIC
  - Network Interface Card
  - Has a unique 12 character Hexadecimal number permanently burned into it at the manufacturer.
  - The number is the MAC Address/Physical address of a computer
- Cabling
  - Twister Pair
  - Fiber Optic
  - Coax Cable

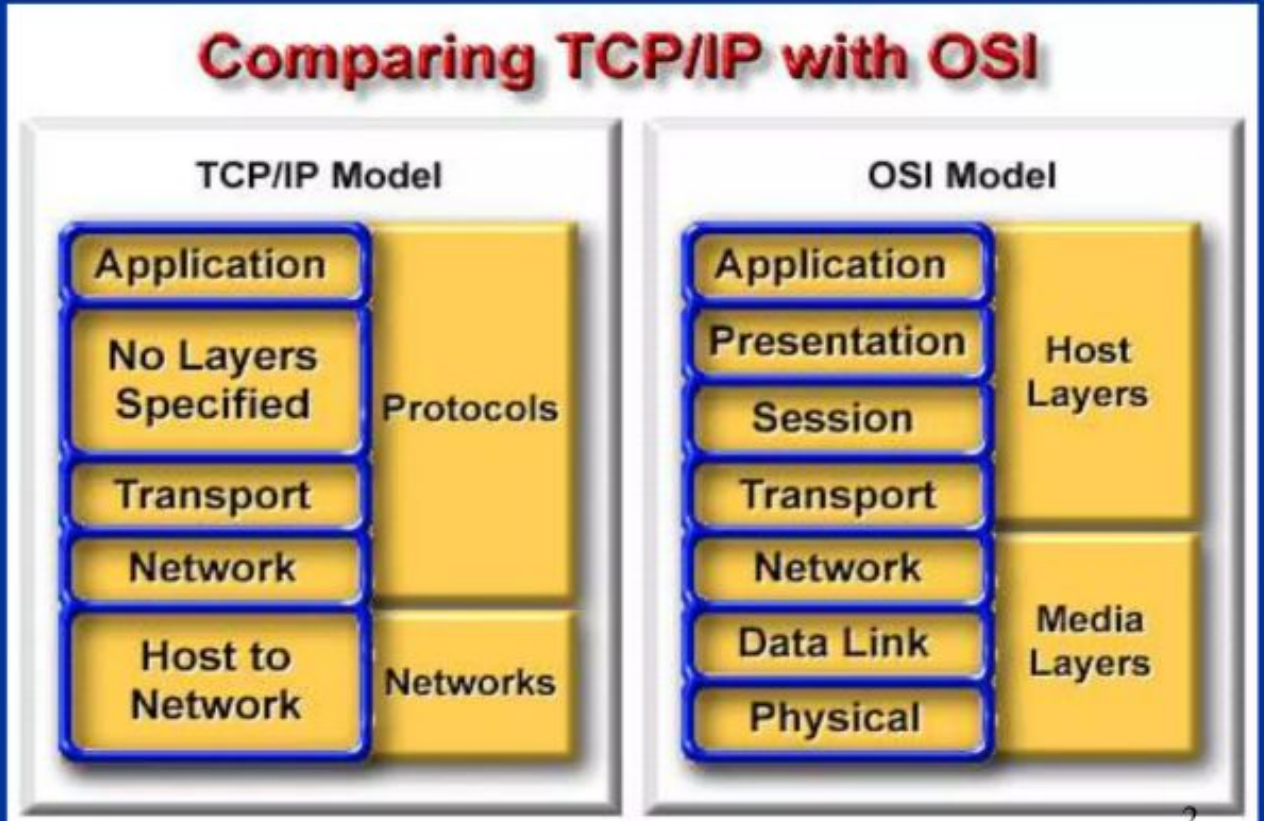
How Does It All Work Together



The TCP/IP Model

- Another Model is the TCP/IP Model.
- There is no universal agreement regarding how to describe TCP/IP with a layered model.
- Most descriptions present three to five layers.
- We use the four layer structure that incorporates the Presentation and Session layers with the Application layer.

Comparing TCP/IP With OSI



The TCP/IP Model (4 Layer)

- Application Layer
  - Interacts with user processes
- Transport Layer
  - TCP guarantees data is received and sent accurately
- Internet Layer
  - IP separates upper layers from the network and manages the connections across the network
- Network Access Layer
  - Incorporates the Network and Physical layers of the OSI model

Remembering the 7 Layers

7 - **A**pplication	**A**ll
6 - **P**resentation	**P**eople
5 - **S**ession	**S**eem
4 - **T**ransport	**T**o
3 - **N**etwork	**N**eed
2 - **D**ata Link	**D**ata
1 - **P**hysical	**P**rocessing

Brief History of the Internet

- 1968 - DARPA (Defense Advanced Research Projects Agency) contracts with BBN (Bolt, Beranek & Newman) to create ARPAnet
- 1970 - First five nodes:
  - UCLA
  - Stanford
  - UC Santa Barbara
  - U of Utah, and
  - BBN
- 1974 - TCP specification by Vint Cerf
- 1984 – On January 1, the Internet with its 1000 hosts converts en masse to using TCP/IP for its messaging

A Brief Summary of the Evolution of the Internet

Memex Conceived 1945

A Mathematical Theory of Communication 1948

Silicon Chip 1958

First Vast Computer Network Envisioned 1962

Packet Switching Invented 1964

Hypertext Invented 1965

ARPANET 1969

TCP/IP Created 1972

Internet Named and Goes TCP/IP 1984

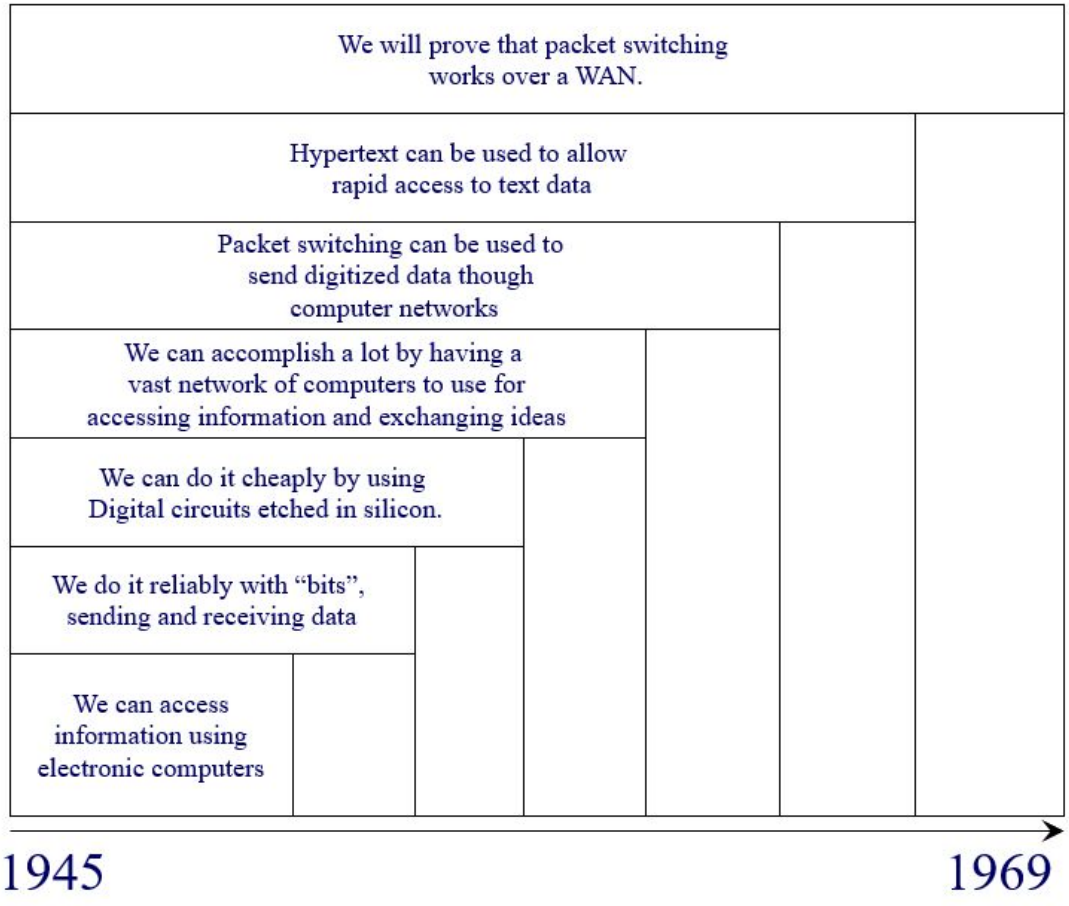
WWW Created 1989

Mosaic Created 1993

Age of eCommerce Begins 1995

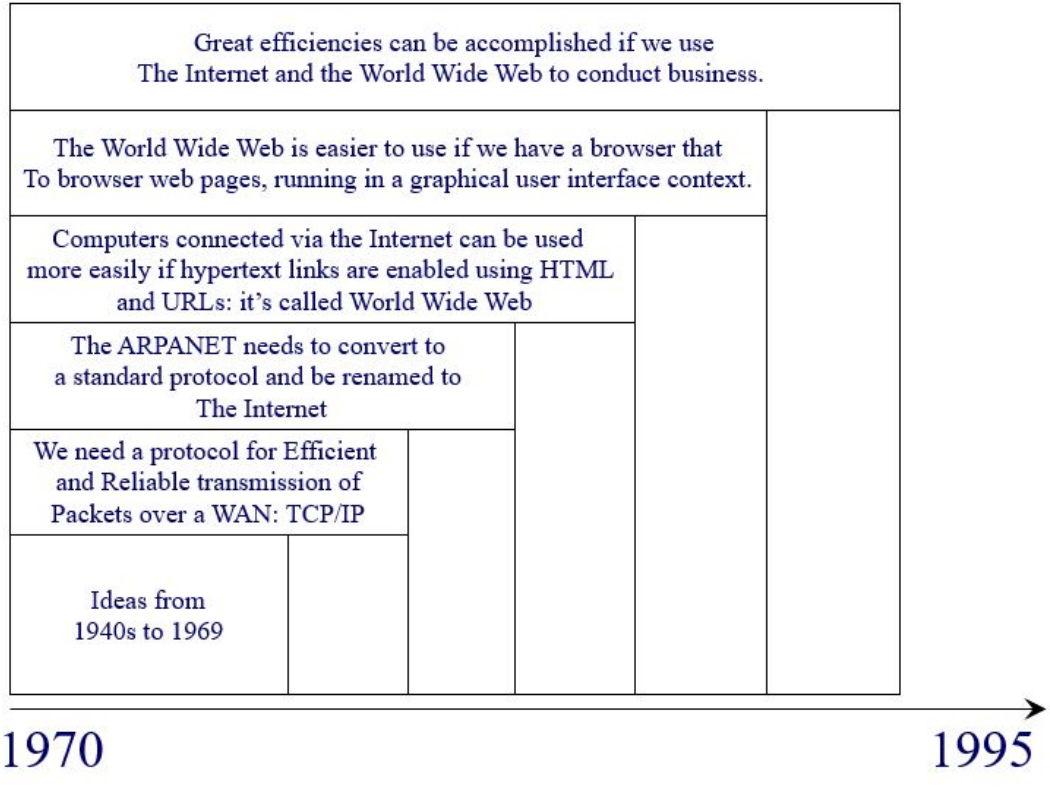
1945 1995

From Simple, But Significant Ideas Bigger Ones Grow 1940s to 1969



Internet Society

From Simple, But Significant Ideas Bigger Ones Grow 1970s to 1995



Internet Society

The Creation of the Internet

- The creation of the Internet solved the following challenges:
  - Basically inventing digital networking as we know it
  - Survivability of an infrastructure to send / receive high-speed electronic messages
  - Reliability of computer messaging

Internet Society

Definition of Internet

- a global computer **network** providing a variety of information and communication facilities, consisting of interconnected networks using standardized communication protocols.
- The Internet, sometimes called simply "the Net," is **a worldwide system of computer networks** -- a network of networks in which users at any one computer can, if they have permission, get information from any other computer (and sometimes talk directly to users at other computers).

Internet thermology

- Internet Technologies is a technical field that covers the necessary skills to develop applications on the Internet or Internet based systems, harnessing e-commerce, cloud, mobile, and Web based technologies.

Common Internet Terminologies

- Browser.
- E-mail
- File transfer protocal (FTP)
- Homepage.
- Hypertext transfer protocol (HTTP)
- Webside.
- Webpages.
- Internet protocol (IP) Address

Browser

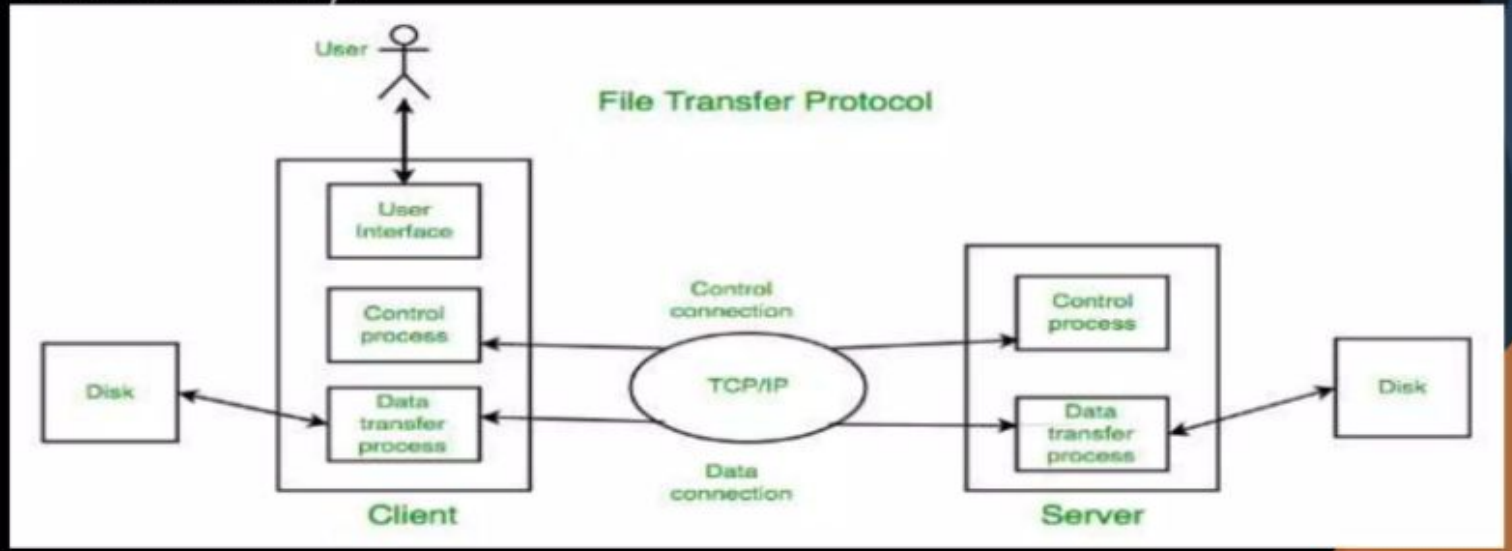
- A piece of software such as Mozilla Firefox and Internet Explorer that allows a computer to access and display documents, view pictures, hear sound, and view video clips from the World Wide Web.

E-mail

Mail that's electronically transmitted by your computer. As opposed to snail mail, e-mail sends your messages instantaneously, anywhere in the world. It has the capability to send messages at any time and to anyone.

File Transfer Protocol (FTP)

The standard method for downloading and uploading files over the Internet. With FTP, you can login to a server and transfer files (meaning you can "send" or "receive" files).



Homepage

The first page that is viewed when the browser starts. It is also the page of a Web site that provides the introduction or content with links.

Hypertext Transfer Protocol (HTTP)

The abbreviation for Hypertext Transfer Protocol. It is the set of rules by which Web pages are transferred across the Internet.

HTTP overview

HTTP: hypertext transfer protocol

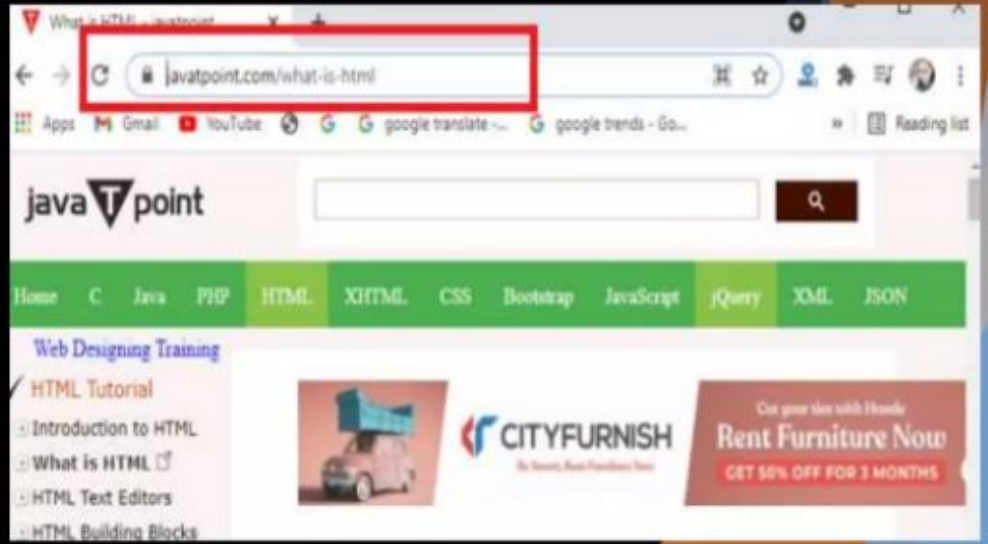
- Web's application layer protocol
- client/server model
  - *client:* browser that requests, receives, (using HTTP protocol) and "displays" Web objects
  - *server:* Web server sends (using HTTP protocol) objects in response to requests

Website

A collection of World Wide Web pages or files.

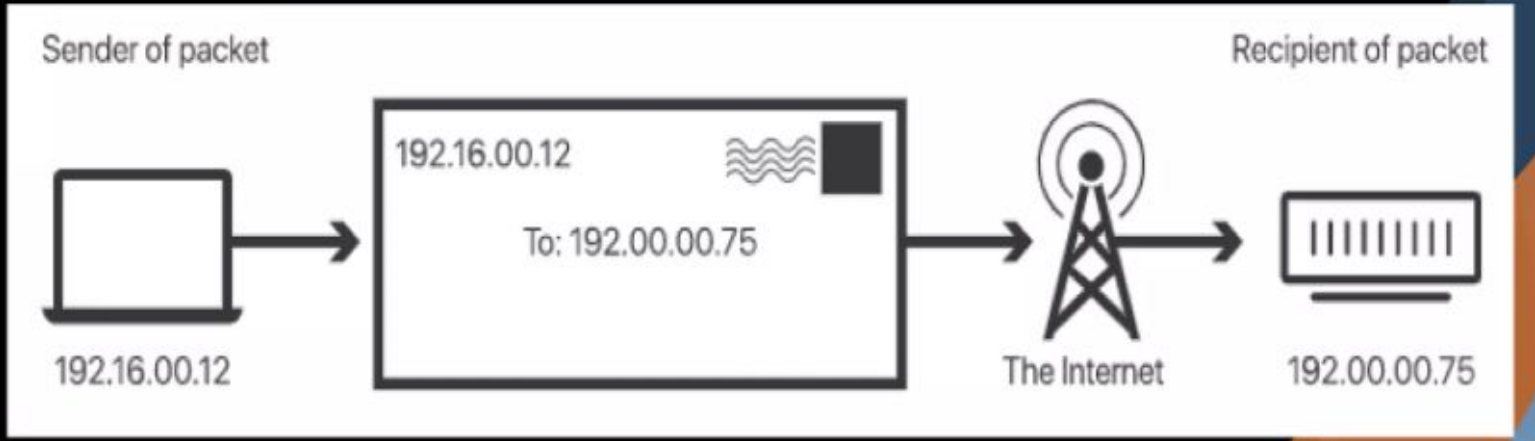
Web Page

A single hypertext file or a page that is part of a Web site.



Internet Protocol (IP) Address

The Internet is composed of local, regional, national, and worldwide computer networks. Each computer on the Internet can be identified by a set of unique numbers that is called an internet protocol (IP) address.



Connecting to the Internet.

Methods available

There are a number of ways of connecting to the internet. These include:

- 'Dial-up'
- Integrated Subscriber Digital Line (ISDN)
- Broadband (ADSL)
- Mobile devices (e.g. latest mobile phones)

Dial-up connections (1)

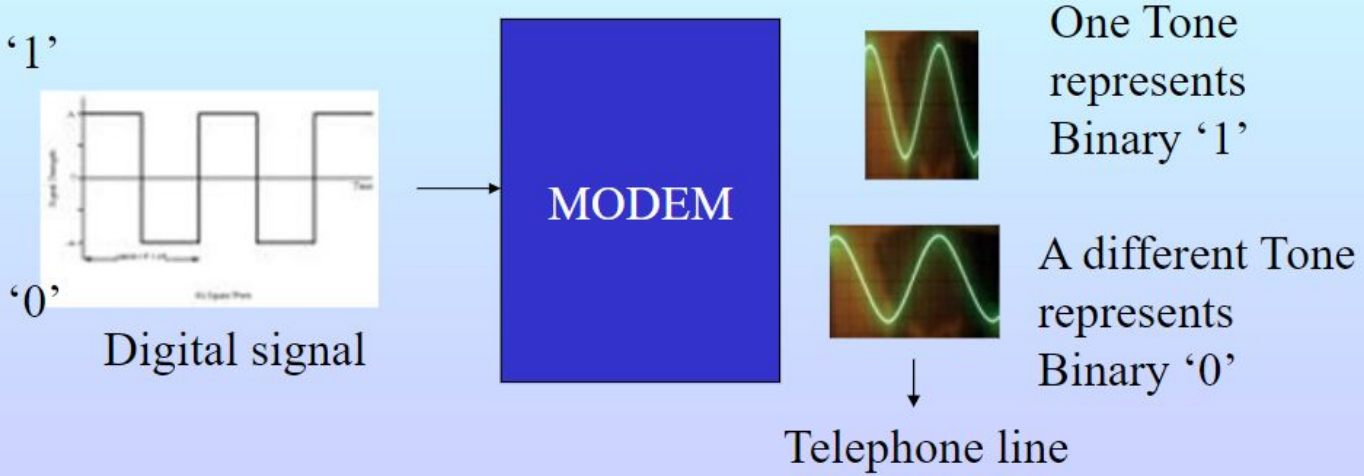
The original telephone system is based on ANALOGUE technology. Analogue signals change smoothly with time. Like this:

Your voice is ‘analogue’ in form – it changes smoothly and so this technology was perfect for carrying telephone conversations.

The trick was how to transmit DIGITAL information over an
ANALOGUE telephone line

Dial-up connections (2)

A device called a ‘Modem’ was developed to allow Digital data to be carried over the analogue telephone system.



This is why dial modems can be heard to ‘whistle’ as they communicate. You are hearing the tones that represent digital information

Copyright www.teach-ict.com

Dial-up connections (3)

- Modems are defined by their speed i.e. how many bits per second can they handle.

- A modern dial-up connection makes use of a 56 Kilobits per second Modem.

- A dial-up connection allows you to connect to the internet via a local server using a standard 56k modem, your PC literally dials (hence the name) a phone number (provided by your ISP) and connects to the server and therefore the internet.

Disadvantages of Dial-up (4)

56Kb is very slow by modern standards. A graphic heavy web page can take tens of seconds to load. WWW even had another joke meaning: the World Wide Wait!

- Dial up ties up your telephone line and so it cannot be used for anything else.
- It takes a while for the computer to set up a connection i.e many seconds.
- Internet Service Providers may charge you by the minute and so a slow link means you pay more.

Advantages of dial-up (5)

Dial-up connections can be very economic and are widely available – anyone with a telephone line can make use of it.

- If you only make occasional use of the internet then it may be the easiest option as all you need is a modem.
- Other technologies such as broadband may not be available in your local area and so 'Dial-up' is the only option.

ISDN

- During the 1980s, a new technology called ISDN or **Integrated Services Digital Network** was developed by BT.
- ISDN makes use of DIGITAL technology.
- It was prohibitively expensive so only businesses could initially afford it. As with all new technology, prices eventually fell and the cost of this technology is now within reach of homeowners.
- For domestic users the BT ISDN service is called 'Home Highway'

ISDN - advantages

An analogue modem allows speeds up to 56 kilobits per second (in theory). Standard ISDN is 64Kb but it allows speeds of up to 128 kilobits per second if your ISP supports it.

- The ISDN service allows you to use the telephone at the same time – it does not tie up the line.

- It is fast to connect – only a second or so.

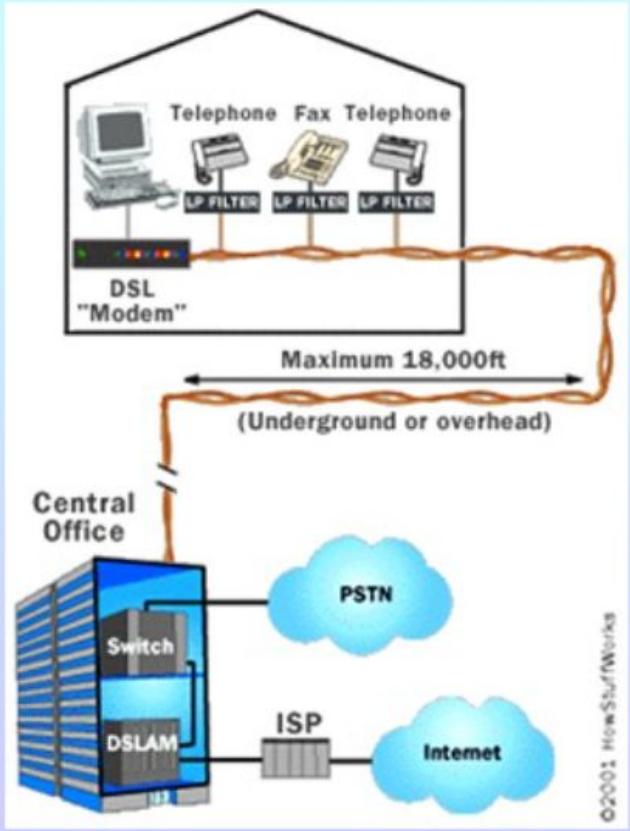
ISDN - disadvantages

- Your main telephone socket has to be changed to an ISDN box by an engineer.
- Your computer has to be fitted with an ISDN modem which is more expensive than a standard modem.
- It is more expensive than Dial-up on a monthly basis.
- Should you move house, you have to pay for your telephone line to be converted back again to a normal line.

What is DSL?

- A new technology came on the scene in recent years called DSL.
- Digital Subscriber Line (DSL) is the generic name for a technology that allows digital data to sent over an ordinary copper telephone line at high speed.
- It is much faster than ISDN. For example in the UK it is common to have a 512Kbs service. Compare this to a dial-up service at 56Kbs or ISDN at 64Kbs
- There are more than one type DSL service. For example
  - ADSL (Asymmetric Digital Subscriber Line)
  - SDSL (Symmetric Digital Subscriber Line)

DSL – Typical setup



What is ADSL?

Providing bandwidth is expensive – it needs special equipment at the exchanges and your telephone line has to be of good quality .

Telephone companies realised that most domestic customers would be downloading far more than they would be uploading. And so it made sense to develop a DSL technology that provided fast download speeds but slower upload speeds.

This service is called Asymmetric Digital Subscriber Line (ADSL) otherwise known ‘Broadband’

ADSL (1)

ADSL connections are becoming more and more widely available and can provide an excellent internet connection.

The connections work by splitting your phone line into two separate channels, one for data (internet) and one for voice (phone calls) which means you can talk on the phone and be connected to the internet at the same time.

ADSL (2)

You will often see ADSL connection services advertised as having different speed specifications, below are some common configurations:

- 256Kbps/128Kbps
- 512Kbps/128Kbps
- 1Mbps/256Kbps

Notice there are two values to each configuration, the first figure states the download speed and the second figure is the maximum upload speed.

ADSL – Typical performance

Average Web Page (50 Kilobytes including images)

- 256kbps download speed - roughly 1.56 seconds to view page
- 512kbps download speed - roughly 0.78 seconds to view page

Average MP3 file (music file) (4 Megabytes)

- 256kbps download speed - roughly 2 minutes and 8 seconds to download.
- 512kbps download speed - roughly 1 minute and 4 seconds to download.

Advantages of ADSL

- Faster downloads compared to dial-up or ISDN
- No need for a second phone line - by allowing voice and data transfer at the same time (you can use the phone as normal while connected to the internet).
- Because ADSL transfers data digitally it doesn't need to convert the data from digital to analogue and back again.
- ADSL connections are ***Always on***, which makes the usual long wait to connect a thing of the past.

Disadvantages of ADSL

- ADSL connections are not available to everyone, you need to be within 3 miles of an ADSL enabled exchange.
- The hardware costs can be quite significant as you will need a special ADSL modem and ADSL filters to use the service, most ISPs allow you to hire these items which can reduce the initial cost.
- Because ADSL connections are ***Always on*** you will need a firewall to protect your PC

APPLICATION OF INTERNET

- Communication
- Job Searches
- Finding books & Study materials
- Health & Medicine
- Travel
- Entertainment
- Shopping
- Stock market updates
- Shopping
- Research
- Business

COMMUNICATION

It is used for sending and receiving messages from one and other through internet by electronic mail
Ex:
Yahoomail.com
Hotmail.com

JOB SEARCHES

Getting information regarding availability of job in different sector and areas.

FINDING BOOKS

Books and other study material stored around the world can be easily located through internet.
Ex:
Encyclopedias

HEALTH AND MEDICINES

People can get various information about various diseases can receive help.

TRAVEL

Can get information about tourist places.

Used for booking holiday tours,hotels,trains
Etc.

ONILNE SHOPPING

By giving our account details perform transactions.

BUSINESS

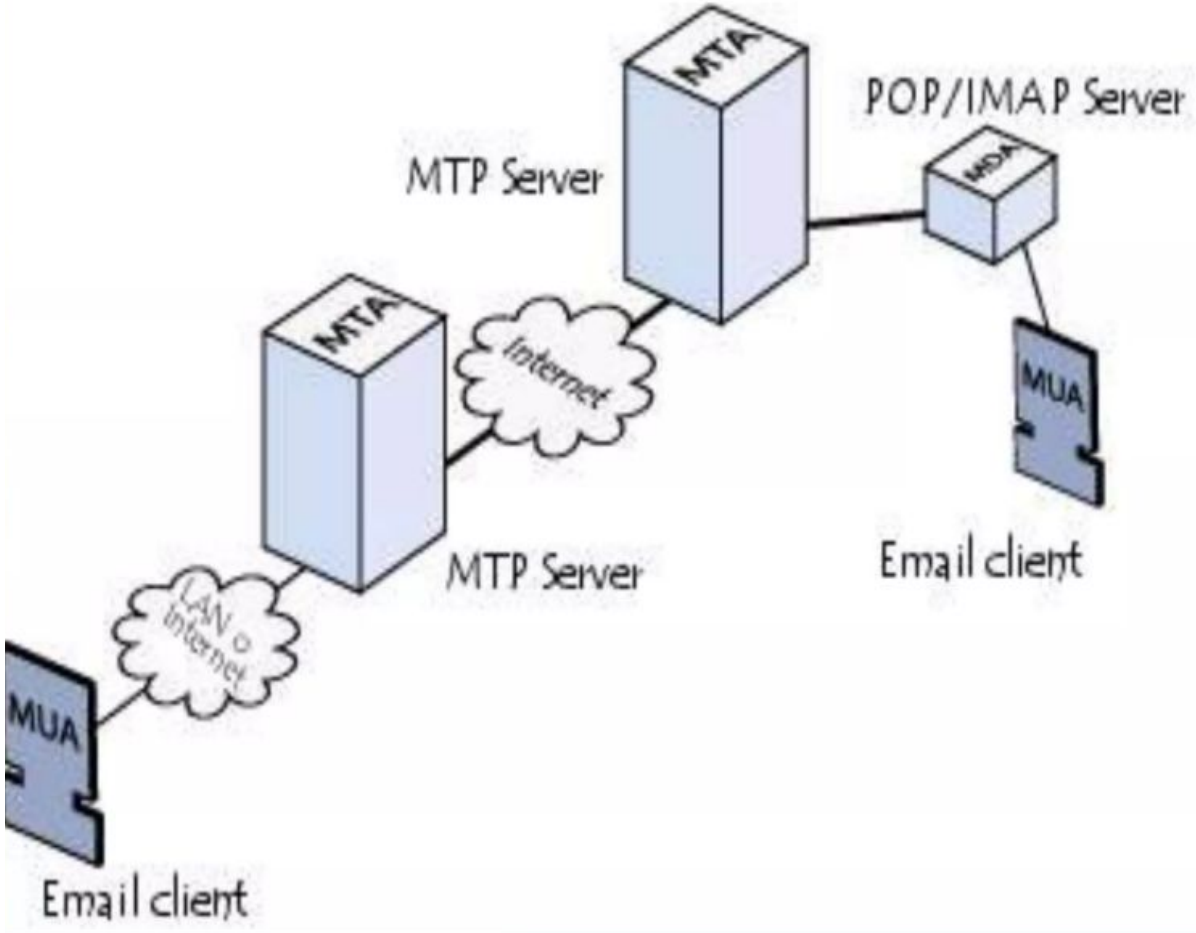
- Information of products can be provided online to the customer.
- Market information.
- Viewing company products.
- Can eliminate middle agencies and can contact with customer directly.
- Provide companies background and financial into website.

How E-Mail Works

- Email is based around the use of electronic mailboxes.
- When an email is sent, the message is routed from server to server, all the way to the recipient's email server.

Mail Transfer Agent (MTA)

- The message is sent to the mail server tasked with transporting emails (called the **MTA**, for *Mail Transport Agent*) to the recipient's MTA.
- On the Internet, MTAs communicate with one another using the protocol SMTP, and so are logically called **SMTP servers** (or sometimes *outgoing mail servers*).
- The recipient's MTA then delivers the email to the incoming mail server (called the **MDA**, for *Mail Delivery Agent*), which stores the email as it waits for the user to accept it.


MTA
POP/IMAP Server
MTP Server
MDA
MTA
Internet
MUA
MTP Server
Email client
LAN o Internet
MUA
Email client

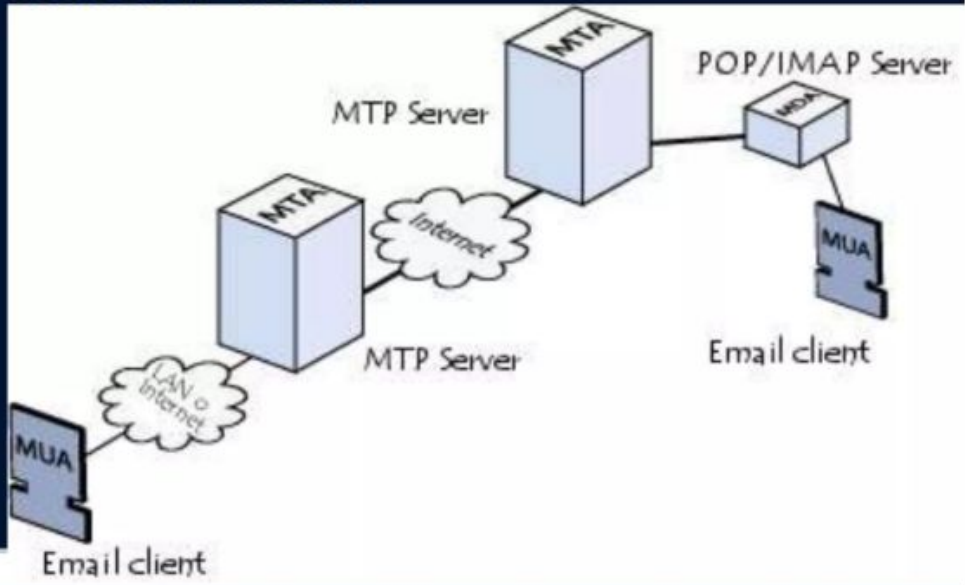
POP3, IMAP

There are two main protocols used for retrieving email on an MDA:

- **POP3 (Post Office Protocol),** the older of the two, which is used for retrieving email and, in certain cases, leaving a copy of it on the server.
- **IMAP (*Internet Message Access Protocol*),** which is used for coordinating the status of emails (read, deleted, moved) across multiple email clients. With IMAP, a copy of every message is saved on the server, so that this synchronization task can be completed.

POP3, IMAP

- For this reason, incoming mail servers are called **POP servers** or **IMAP servers**, depending on which protocol is used.



Difference Between MTAs & MDAs

- To use a real-world analogy, MTAs act as the post office (the sorting area and mail carrier, which handle message transportation), while
- MDAs act as mailboxes, which store messages (as much as their volume will allow) until the recipients check the box. This means that it is not necessary for recipients to be connected in order for them to be sent email.

E-Mail Security

- To keep everyone from checking other users' emails, MDA is protected by a user name called a **login** and by a **password**.
- Retrieving mail is done using a software program called an **MUA** (*Mail User Agent*).
- When the MUA is a program installed on the user's system, it is called an **email client** (such as Mozilla Thunderbird, Microsoft Outlook, Eudora Mail, Incredimail or Lotus Notes).
- When it is a web interface used for interacting with the incoming mail server, it is called **webmail**.

Search Engines

- Information retrieval tools
- Identify search topics, software finds matches
- Not “real time” search of Internet
- Search engines create index of Web
- Size of index varies

Popular Search Tools

- Google (www.google.com)
- Yahoo (www.yahoo.com)
- Ask Jeeves (www.askjeeves.com)
- AllTheWeb.com (www.alltheweb.com)
- HotBot (www.hotbot.com)
- Teoma (www.teoma.com)
- AltaVista (www.altavista.com)
- Lycos (www.lycos.com)

Most Search Engines:

- Have forms to enter keywords
- Links to help & advanced search tools
- Subject categories or directories

Simple Searching

- Keywords – terms relating to subject
- Spelling counts
- Enter keyword in search engine and click “SEARCH”
- Results can vary
- Meaning may depend on context

Advanced Searching

- Check HELP or ADVANCE SEARCH options
- AND: all keywords must be present
- OR: any of keywords can be present.
- NOT: excludes specified keywords

Phrase Searching

- Check HELP or ADVANCE SEARCH options
- Enclose in quotation marks (“”)
- Matches phrase EXACTLY
- Also called “literal”

Natural Language

- Check HELP or ADVANCE SEARCH options
- Search with sentences/ phrases used in conversation
- Search engine tries to determine keywords
- Be careful, results can very widely

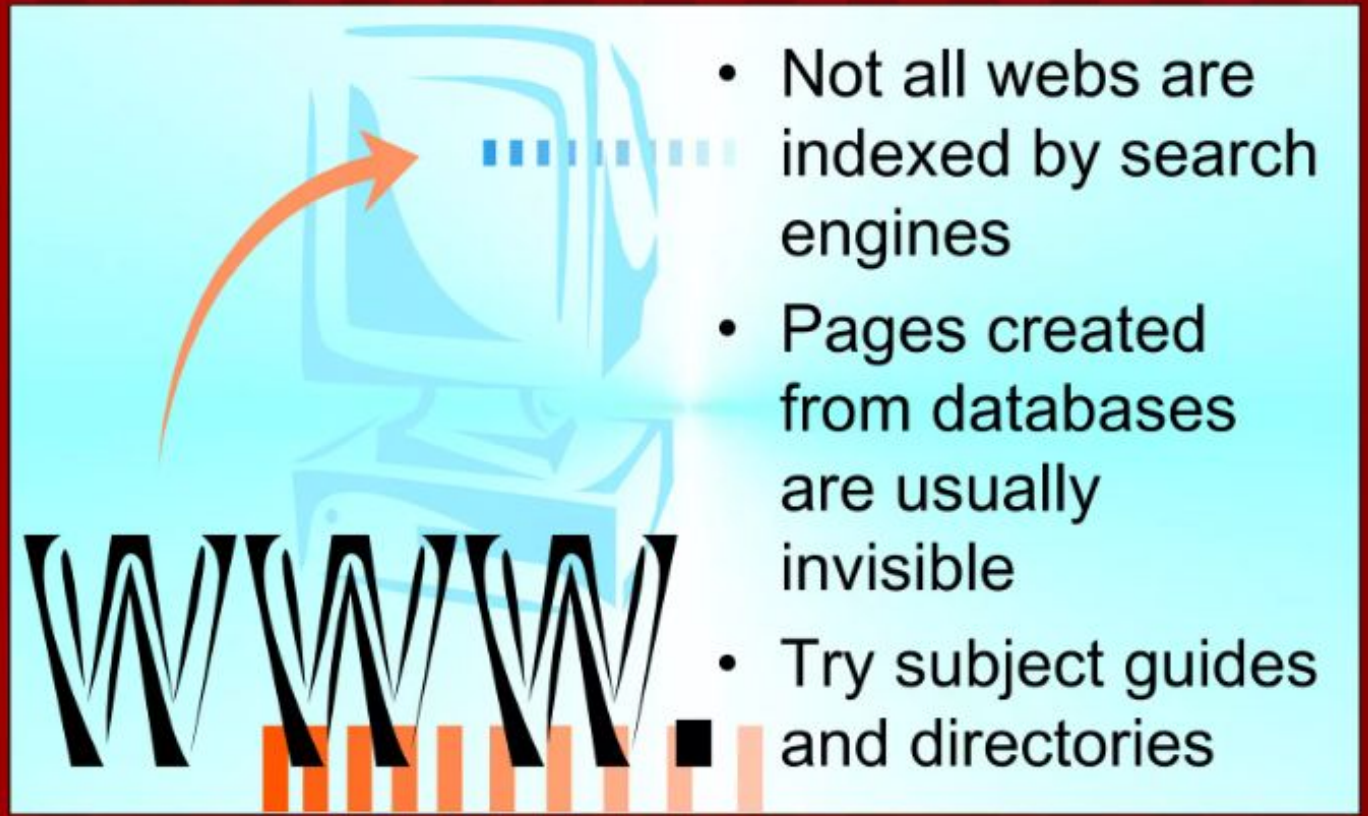
Meta Searching

- Enter keywords in 1 dialog box
- Automatically search many different search engines
- Cannot use each search engines unique advanced features

Meta Search Engines

- Clusty (clusty.com)
- Copernic Agent (www.copernic.com)
- Dogpile (www.dogpile.com)
- KartOO (www.kartoo.com)
- Mama Metasearch (www.mamma.com)
- MetaCrawler (www.metacrawler.com)
- SurfWax (www.surfwax.com)

Invisible Web



- Not all webs are indexed by search engines
- Pages created from databases are usually invisible
- Try subject guides and directories

Invisible Web Guides

- Academic Info (www.academicinfo.net)
- Free Pint (www.freepint.com)
- Infomine (www.infomine.ucr.edu)
- Librarians' Index to the Internet (www.lii.org)
- ResearchBuzz (www.researchbuzz.org)
- The Scout Report (www.scout.cs.wisc.edu)

Coding Languages of Internet

Client Side Programming

- Executed at the client machine
- Interacts with the user
- Languages:
  - HTML
  - CSS
  - Java Script
  - Php
  - XML

The Coding Wizards
Code Wiz Academy

Structure of Web Page

- HTML- Hyper Text Markup Language
- Head
- Body

Design of Web Page

- CSS- Cascading Style Sheet
- Stylising the layout
  - Font
  - Color
  - Spacing
- Device Specific

The Coding Wizards

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Web Page

- Code is interpreted line by line
- Interpreter is pre-built in web browser
- CSS code is linked/embedded in HTML file
- Code as specified in HTML file is executed

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Steps

- Write the structure of web page
  - Using HTML
- Add designing to web page layout
  - Using CSS
- Rethink the layout and amend the design as required

Saving

- Create a separate folder for code
- Save all the files HTML, CSS, Media in same folder or sub folder within it
- While linking external CSS files specify the path to the files

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Editor

- Online Editor
  - http://scratchpad.io/oafish-taste-6473
- Offline Editor
  - Windows OS
    - Notepad++
  - MAC OS
    - Brackets

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JS

- Java Script
- Client Side scripting
- Code is written as plan script
- Executed on client machine- does not need compilation
- Executed by Java Script Engine in web browser
- Dynamic user interaction

```
<script language="javascript">
        alert( 'Hello, world!' );
</script>
```

XML

- Extensible Markup Language
- Defines set of rules for format of document
- Used to store and transport data
- Define your own tags

```
<garden>
    <soil>Organic</soil>
    <water>Daily</water>
    <manure>Monthly</manure>
    <body>Love your plants they love you back!
</body>
</garden>
```

Php

- *Hypertext Pre-processor*
- Open source general-purpose scripting language
- Can be embedded into HTML
- Code is executed on the server
- HTML code is sent to the client

```
<html>
    <body>
        <?php
            echo "Hello World!";
        ?>
    </body>
</html>
```

Code Wiz Academy

What is computer Virus?

* A computer virus is a software program with a ability to make copies of itself ,which attacks itself to other applications or files.

Types of computer Virus.

- There are three types of computer virus
  - *Boot virus*
  - Program virus
  - Micro virus

BOOT VIRUS:

- Boot viruses infects the boot records of hard disk and floppy disks. Generally these are considered more dangerous than program viruses.

Program Virus

- These viruses infect program files, which have extensions such as .COM,.EXE.,SYS.,DLL,.OVL, or .SCR. program files are attractive targets for computer virus because they are widely used and have relativity, simple formats to which viruses can attach themselves.

Macro Viruses

- These viruses infects data files. For example, Microsoft word document and template files are susceptible to macro virus attacks they spread rapidly as infected documents are shared on networks or downloaded from internet sites.

What a computer virus do?

- A virus tries to take control of computer system it has infected at the first opportunity availible.it makes copies of itself and also try to carry harmful task written in its program. This process can happen so quickly that the user is not even aware of the presence of a virus in his/her computer.

Top 5 Sources of Computer Virus Attack

- The most potent and vulnerable threat of computer users is virus attacks. Virus attacks hampers important work involved with data and documents. It is imperative for every computer user to be aware about the software and programs that can help to protect the personal computers from attacks. One must take every possible measure in order to keep the computer systems free from virus attacks. The top sources of virus attacks are highlighted below:
  - Downloadable Programs
  - Cracked Software
  - Email Attachments
  - Internet
  - Booting From CD

Best antivirus software.

- Norton antivirus.
- MacAfee virus scan
- Kaspersky antivirus

BEST OF LUCK